Decoding Population Dynamics: A Multifaceted Analysis of Growth Trends in India, China, U K, U.S.A, & Brazil

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Submitted by

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Abstract

Demographic patterns are key to understanding global population shifts. This research examines population growth trajectories of India, China, the UK, the USA, and Brazil from 1950 to 2023, drawing data from the United Nations Department of Economic and Social Affairs, Population Division. Using statistical analysis, the study identifies influential variables for these nations' populations and their worldwide impact. Methods such as Linear Regression and ARIMA predict the Crude Birth Rate (CBR). For overall population projections, Linear Regression, aided by the Prophet method, is utilized. The findings highlight CBR, Crude Death Rate (CDR), and Net Migration Rate (CNMR) as pivotal in guiding population trends. It points to a rising young populace in India and Brazil, an aging demographic in China due to its one-child policy, and the UK's aging pattern. Migration trends show the global appeal of the USA and UK, with India and China emerging as key centers for human capital distribution. The role of government policies in directing demographic changes is clear. Linear Regression's efficacy in forecasting CBR, CDR, and total population is confirmed, with the Prophet method excelling in broader population predictions. Historical events, like World War I, and policies such as China's one-child rule, influence demographic changes. Alterations in populous countries like India and China impact global sectors, including business and diplomacy. With their geopolitical and economic significance, the UK and the USA play a vital role in global demographic trends. In conclusion, this research provides insights into global demographic factors, highlighting the significant role of these countries in worldwide population dynamics, serving as a resource for policymakers, scholars, and industry professionals.

Acknowledgement

I would like to express my heartfelt gratitude to my esteemed supervisor, Dr.Philip Trevelyan, for his consistent support and guidance throughout this dissertation. I am really thankful for your valuable feedback, encouragement, and patience. His steady commitment has been crucial in the successful completion of this dissertation.

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I wish to acknowledge the grace and blessings of the Almighty throughout the study.

Chapter 1

1.1 Introduction

'A population is any large group that shares at least one trait. People are only one part of a population. People, animals, businesses, buildings, farms, goods, or events are all examples of populations. [of Statistics, 2023]'.Demography, the study of populations, primarily concentrates on births, migration, and aging, (including death). [Max Planck Institute for Demographic Research, n.d.]. While a detailed analysis of every individual would offer the most accurate results, practical constraints often make this method impractical, especially in large populations. Instead, demographers often rely on two main methods: a complete enumeration, known as a census, and a partial enumeration, termed as sampling. A census provides a complete overview of a population, such as the households and occupation of an individual. On the other hand, sampling involves selecting a representative group from a larger population, serving as an analytical representation of the whole. Both methods offer valuable insights about the population. Several factors, including the scope of the research endeavor, the resources available, and the required level of accuracy, are taken into consideration on deciding whether to conduct a census or utilize a sampling strategy [of Statistics, 2023].

Surveys and polls are not just data collection tools, they are vital for uncovering broader societal trends and specific challenges. Organizations like the Centers for Disease Control and Prevention enhance our collective understanding of populations through ongoing research initiatives [Momoh, 2021]. The process of quantifying a population usually takes place annually, often on the first day of January, and is heavily influenced by the most recent decennial census data. This census data is used by the government to understand the composition of their citizenry, including variables such as age, ethnicity, gender, income, and occupation. This information is not only crucial for tax collection but also plays a significant role in the allocation of resources to various social programs and infrastructure projects. Hence, the study of demography becomes indispensable in guiding public policy and business decisions. The desire of higher living standards which includes appropriate nutrition, healthcare, literacy, educational access, and employment opportunities is a common desire. A prominent constraint to the achievement of these goals is the existing rate of population growth where Population growth is determined by the rise in the number of people in a population in a given year [IGI Global, n.d.]. 'The prevailing global population of 8 billion is expected to double in 2057. If current rates of growth continue, the population will peak around 10.4 billion in 2086' [Ritchie, 2022].

Such exponential population growth puts a disproportionate load on efforts targeted at economic progress, stifling programs aimed at improving human welfare. Furthermore, due to the linked structure of global systems, this issue surpasses national borders, appearing as a global challenge [National Academies of Sciences and Medicine, n.d.]. As the world's population approached 8 billion, even modest consumption strains the planet's ecological capability. Current average consumption exceeds sustainable levels by more than 50 percent. The term "overpopulation" refers to the Earth's incapacity to replenish the annual resource consumption of its inhabitants. Since 1970, this has been a rising source of concern, with each year aggravating the ecological imbalance. Understanding the causes of overpopulation and overconsumption is critical because they affect everything from climate change to geopolitical stability [Bish, 2020].

In order to offer informed recommendations concerning global population trends, a rigorous examination of population growth strategies in the most populous nations is essential. The objective of this thesis is to delve into the fascinating world of population growth, focusing on five key countries that play a significant role on the global stage: India, China, the UK, the USA, and Brazil, with China and India taking the lead in global population, followed by the USA. Furthermore, Brazil distinguishes itself as the fifth-largest country by land area [Kids, 2021] and ranks seventh in terms of its population as of July 2023 Population [2023]. The UK was chosen due to a noticeable escalation in population figures has been observed, primarily attributed to an increase in immigration since the year 2000. By examining the salient factors influencing population growth and assessing the consequences of these demographic shifts on a global scale, a more subtle understanding of our societal environment can be achieved. Such insights are indispensable for the formulation and implementation of sustainable practices and policies designed to address the challenges presented by these demographic changes.

1.2 Objective and research questions

The principal aim of this dissertation is to engage in a rigorous analysis of population growth trajectories within a selected cohort of nations. The research questions are:

- What variables are most influential in determining the population growth rates within the targeted nations?
- How do variations in population growth rates, either escalation or diminution, impact overall population size and structure?

- What is the cumulative effect of individual country population growth trends on the global demographic composition?

1.3 Literature Review

Population growth continues to be a subject of immense global importance, with implications ranging from economic development to sustainability. Given the significance and complex dynamics of population growth, predictive modeling has emerged as a crucial tool for demographers and policymakers alike. This literature review aims to explore past research focused on identifying variables affecting population growth, the methodologies employed for predictive analysis, and particularly the metrics that are crucial for understanding population dynamics. An exhaustive understanding of variables affecting population growth is essential for accurate predictive modeling.

[Lee, 2003] offers a fundamental framework through *the demographic transition model*, which covers key factors such as birth rates, death rates, and migration patterns. This model functions as a pivotal element for classifying the developmental stages of the countries under scrutiny and subsequently shapes their unique population growth trajectories. [Lee, 2003] integrates various theoretical paradigms, including Malthusian, neoclassical, and new household economics, highlighting the necessity for a multi-dimensional approach to grasp the complexities of population growth. The Malthusian theory posits that population grows geometrically, while food supply increases arithmetically, thereby issuing a warning against the risks of uncontrolled population growth. Concurrently, [Popkin, 1998] investigates the impact of healthcare accessibility and lifestyle alterations on birth rates, while [Bongaarts, 2009] concentrates on fertility rates, migration, and mortality as pivotal variables. [Popkin, 1998] discusses some of the potential health consequences of significant changes in nutrition, activity, and body composition occurring in low- and middle-income countries. Sudden changes in food structure, particularly those associated with urbanization and occupation, were discovered to alter both fertility and birth rates. [Bongaarts, 2009] addresses that the global demographic landscape is undergoing quick transformation, including a massive population expansion, with about four billion people added since 1950 and anticipated outcomes indicate a split, denoting possible stagnation in developed regions and continuous growth in developing countries. Significant changes in fertility rates and life expectancy are occurring in accordance with this population increase. These trends have resulted in younger populations in high-fertility countries and aging demographics in developed countries. Similarly, this article examines the period from 1950 to 2050, reflecting the most

rapid phase of demographic development thereby providing historical data to our research. Thus, these studies align closely with our objective of identifying principal variables.

Additionally, [Lee, 2003] introduces the concept of the "demographic dividend", suggesting that nations with declining fertility rates could experience economic advantages owing to a larger workforce. This idea finds empirical support in studies by [Galor and Weil, 1999], among others. Complementing this, [Bongaarts, 2009] clarifies the concept of "population momentum," explaining that a population may continue to grow even when fertility rates have reached equilibrium. This phenomenon is attributed to a youthful demographic that has yet to reach reproductive age, and consequently, their eventual participation in reproduction would lead to a population surge. The idea of population momentum serves as an analytical lens for comprehending why specific countries may persist in witnessing population expansion despite diminishing fertility indices. These works collectively provide a multifaceted understanding of population growth, enriched by diverse theories and methodologies. [Lee, 2003] defines the demographic transition as a fundamental shift from high to low rates of birth and death, typically accompanying socio-economic development. This transition affects not only the rate of population growth but also the age distribution within the population, with societal complications in healthcare, pensions, and workforce availability. [Bongaarts, 2009] further explores how fertility, mortality, and migration patterns collectively shape this transition and thereby influence population trends.

[Gu et al., 2021] further extend this understanding by discussing the regional and country-specific trends in population growth. The metrics employed in the study of population dynamics have undergone significant transformation to include more refined indicators beyond merely evaluating total population. One of the pivotal studies in this domain, [Manchanda et al., 2014], explored the metric of life expectancy to comprehend the role of healthcare in affecting population dynamics. Their work articulated that life expectancy serves as an insightful indicator for capturing the overall well-being of a population, thereby aligning well with the focus of dissertation on metrics such as life expectancy, CBR, and CDR. [Gu et al., 2021] offered an intricate analysis of global demographic shifts, emphasizing the variability in growth rates across different regions. This study is significant as it aids in understanding the heterogeneity in metrics like population growth, population density and sex ratios across nations, including India, China, the UK, the USA, and Brazil. [Gu et al., 2021] offers an extensive examination of global population dynamics. Emphasizing the variability in growth rates across regions, the study underscores the impact of fertility rates, aging populations, and migration

patterns. on societal and economic parameters. Rapid population growth brings along a host of socio-economic challenges, including, but not limited to, resource allocation, infrastructure, and social welfare. [Sadigov, 2022] specifically discusses the mismatch between rapid population growth and socioeconomic stability .

[Amonker, 1974] provided a detailed analysis of India's complex demographic environment, with a focus on essential characteristics such as population growth, distribution, and composition. Crucial procedures and the effectiveness of family planning initiatives was also investigated. India has a profound population scale, characterized by rapid expansion and congestion. While death rates have decreased as a result of advances in healthcare and agriculture, fertility rates remain high, putting further strain on land and resources. Despite successive *'Five-Year Plans'* from 1951 focused on population control, family planning programs have had minimal success, indicating the need for a more diversified strategy [Chandna, 1989]. The study emphasizes the complexities of India's demographic challenges, both quantitative and qualitative. These findings are an asset for our understanding of Indian demography.

[Zhang, 2017] reviews the pioneering one-child policy of China established in 1979, that disciplines families who has surpassed the conception limit. The policy's impacts on family structures are complicated because of current family planning policies and concurrent economic reforms, both of which have previously altered fertility rates. An examination of this literature demonstrates the policy's effect on fertility as well as prospective alterations in human capital investment. Preliminary data imply that divorce rates, workforce participation rates, and migratory patterns are affected. [Cao et al., 2015] investigated the policy's implications for family-run enterprises, especially regarding succession planning, which have indirectly influenced demographic expansion. [Feng et al., 2016] critique the policy as an inadequate public measure, emphasizing the urgency to tackle issues arising from diminished fertility rates and analyze health-related concerns in the context of the two-child policy. These contributions offer a multi-layered understanding of national-regulated population management in China.

[Boiko, 2020] examines the evolution of migration policy of the UK from 1990 to 2020, illustrating the country's complex interaction with globalization trends. The UK, known for its high living standards and wages, continues to attract a diverse migrant population, particularly from post-colonial countries and Eastern Europe. Prior to its departure from the European Union in 2019, the UK has shown substantial independence in migration regulation, particularly when contrasted to its European Community acquaintances. This autonomy was most visible in the

policy focussed on national security, which was established during Gordon Brown's administration (2007-2010). Significant legal reforms were implemented under John Major's Conservative reign (1990-1997), with a focus on introducing quota-based migration flows.

[Lomax et al., 2019] reveals that the UK is undergoing demographic uncertainty due to its separation from the European Union (Brexit). Four projection scenarios were used to estimate future ethnic diversity. Based on the importance of international migration in their population growth and existing demographic momentum, these scenarios are expected to have an uneven influence on ethnic groupings. British and Irish populations are predicted to fall and migration growth rates of other ethnic groups such as people from Africa, Pakistan and Bangladesh are likely to increase.

[Hirschman, 2014] identified that around 13% of the population of the USA is foreign-born, and if youngsters of the foreign-born are included, approximately one in every four Americans is a member of the recent immigrant group. Despite existing biases, substantial data suggests that this population benefits the American economy as well as society. There is little evidence that such immigration has a negative influence on native-born salaries or employment, and immigrants are found in a variety of scientific and cultural disciplines. Hence, Migration emerges as another pivotal variable that can substantially alter both the size and the demographic composition of a population. An extensive review by [Hagen-Zanker, 2008] identifies various catalysts for migration, from economic motivations to political instabilities. Further specificity is provided by [Savaş et al., 2023], who explore international retirement migration, highlighting the demographic-specific causes and consequences. In predictive models, understanding the influence of migration is critical, which might capture these long-term shifts more effectively. Given the multi-dimensional nature of population growth, various predictive modeling techniques have been employed in the field.

[Otoom, 2021] compared the performance of 17 different machine learning models including popular models such as Linear Regression, Time Series Analysis and Random Forest, to shed light on their efficacy in the context of population prediction and among the algorithms evaluated, the Random Forest technique emerges as particularly effective, especially when grappling with data sets that have gaps. Such insights are invaluable research, endorsing the applicability and efficiency of machine learning techniques in demographic predictions. [Heyde, 1986] found that even non-stationary population models in the demographic studies, can be stated in an auto-regressive format using random coefficients. The study points out that approaches devel-

oped for stationary time series are not only useful, but also directly applicable to population dynamics analysis. [Natekin and Knoll, 2013] has further delved into another machine learning algorithm, Gradient Boosting which excels in forecasting intricate, non-linear associations. In addition to that, this model is very adaptable to the specific needs of the application, such as learning with regard to different loss functions.

In summary, the literature serves as an essential theoretical and methodological foundation for our dissertation, fortifying our goal of expanding the existing scholarly work through a diverse array of predictive models focused on population growth. This research aims to offer a thorough understanding of the intricate interactions between crucial variables identified in this review such as CBR, CDR, life expectancy, migration rates, and population sex ratio, and their widespread impacts across multiple nations, including but not limited to Brazil, China, India, the United States, and the United Kingdom. Also, it is understood that influential frameworks like the demographic transition model, healthcare accessibility, and lifestyle changes have a significant impact on population growth rates. Our research agenda also involves applying these machine learning models to our data set for meticulous examination and subsequent selection of the most effective predictive model. In addition to traditional machine learning methods, we plan to utilize the Prophet, an advanced forecasting tool created by "Facebook's Core Data Science team" [Prophet, 2019]. Prophet excels in time-series data forecasting and is notably effective for handling data sets with several seasons of historical data. It also offers the advantage of resilience to missing data and sudden shifts in trends, thereby demonstrating its adeptness in managing outliers [Prophet, 2019]. In chapter 2, the data set undergoes a cleaning process, followed by an exposition of its statistical characteristics. Chapter 3 consists of an array of methodologies applied for data prediction and forecasting. The methods included in Chapter 3 are then applied to the data set from Chapter 2 and the results obtained are discussed in Chapter 4. Finally, Chapter 5 holds the conclusions derived and presents future work.

Chapter 2 : Data Exploration

In the domain of academic research, the definition of "population" varies based on the specific field [Momoh, 2021]. While the sampling method is one of the methods employed to analyze population characteristics, such techniques become superfluous when inclusive data is available. This dissertation uses a complete data set to analyze population traits of the chosen five countries from 1950 to 2023 which eliminates the need for sampling methods. The study evaluates populations using multiple statistical measures, aiming to understand changes in population, its density, and demographic composition. These yearly statistics consider natural demographic changes and net migration. Natural demographic shifts result from the difference between annual births and deaths, while net migration assesses both immigrants (incoming individuals) and emigrants (outgoing individuals). The CDR and CBR are pivotal indicators of population. Defined annually, CDR and CBR represent the number of deaths and births per 1,000 individuals. Another key metric is life expectancy, estimating the years a newborn might live based on current mortality rates while population density indicates individuals per unit area and it changes due to factors like birth rate, death rate, immigration, and emigration. For instance, rising birth rates and immigration enhance population density, while rising mortality and emigration reduce it [ec.europa.eu, n.d.]. Another significant metric is the sex ratio, indicating the male-to-female ratio within a population for a specific year. A biased sex ratio (1:1) can hinder reproduction and, consequently, influence population growth trajectories, while a declining population might show a female-biased ratio, aiding in population recovery [Tarsi and Tuff, 2012].

2.1 Data collection and cleaning

The data for our study has been sourced from the official website of the *"United Nations, Department of Economic and Social Affairs Population Division"*, [Nations, 2022] which provides information on the world population as well as the population of our selected five countries. The data spans from 1950 to 2100 with actual data from 1950-2022 and predicted data from 2023-2100. The initial data set of global population consists of 67 columns and 43,472 entries and has no null or missing values so very little data cleaning was required. In this research, we meticulously extracted particular rows and columns that captured key indicators, identified from the literature review. The final data set contains 345 rows and 33 columns (described in Appendix 1). Prior to the statistical analysis, it is pertinent to scrutinize a set of graphical depictions portraying the heterogeneous population dynamics across the chosen quintet of nations. The graphs for these visualizations have been sourced from a reputable authority,

namely the *"Department of Economic and Social Affairs Population Division of the United Nations"* [Nations, 2019]. In addition to these plots, this research inquires into an analysis of variables pinpointed from the literature review, including CBR, CDR, Population growth rate, Crude Net Migration Rate, and total population. These variables are examined for our chosen five countries over a time period from 1950 to 2023. The statistical exploration commences with the population pyramids of the selected five nations, emphasizing the distribution of age and gender over a period spanning from 1912 to 2022. In population research, a population pyramid frequently serves as a key analytical tool that visualises individuals by age groups for a specific year [ec.europa.eu, n.d.]. This display usually situates the oldest age brackets at the top and progresses downward to the youngest. The pyramid traditionally segregates men to the left and women to the right. Researchers in the field of population ecology utilize this tool to quantitatively assess demographic parameters. These parameters include several key indicators, such as the size of the population, its density, age structure, birth rates, mortality rates, and population sex ratio which could shape the dynamics of population growth or decline. Figures 1 and 2 illustrate the population pyramids of India and Brazil.

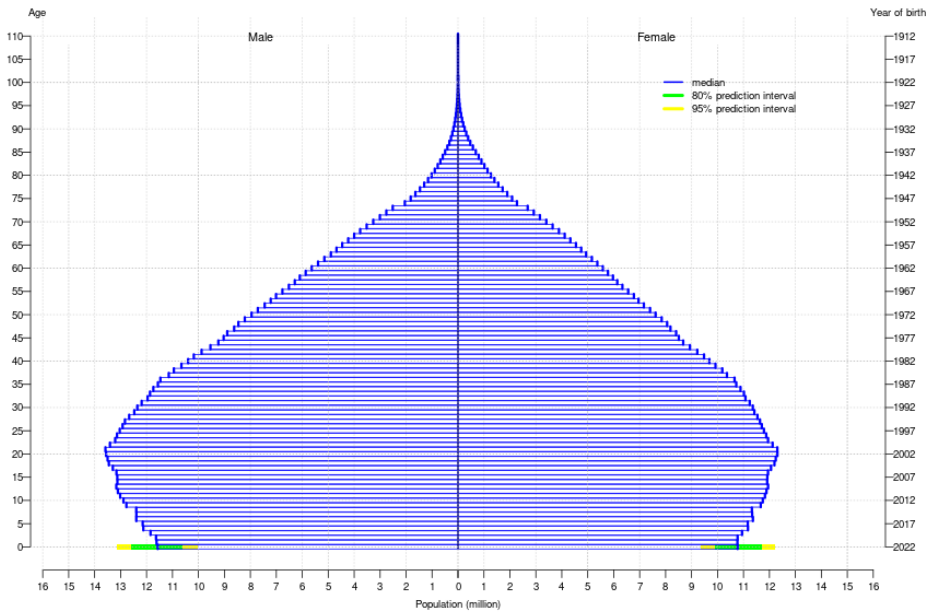


Figure 1: Population of India by age and sex in 2023

Both nations exhibit largely analogous population pyramids; however, they exhibit a distinct divergence in the demographic segment comprising individuals aged 80 to 95 years. Unlike India, where the gender distribution among the elderly appears to be balanced, Brazil, along with other nations, exhibits a demographic skewed toward females in this age group. This gender equilibrium in the elderly population of India has been a consistent indicator of societal

health since the early 20th century. Figure 1 shows that the apex of India's demographic curve was reached in 2001, and Figure 2 demonstrates that Brazil accomplished its highest birth rate earlier, particularly in 1984. After these respective peaks, both countries experience a decline in population without any further significant growth.

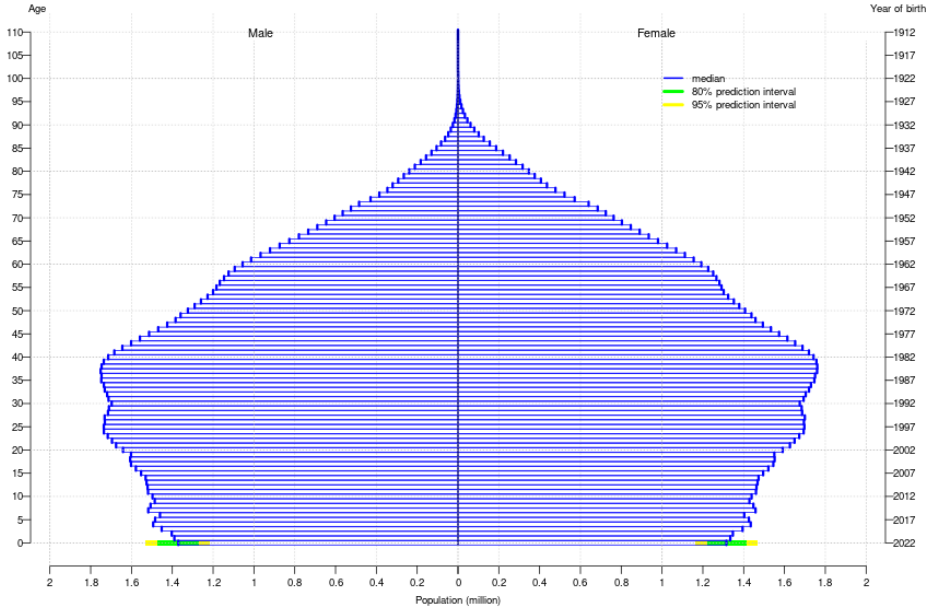


Figure 2: Population of Brazil by age and sex in 2023

Similarly, Figures 3 and 4 depict the population pyramids of the UK and the USA.

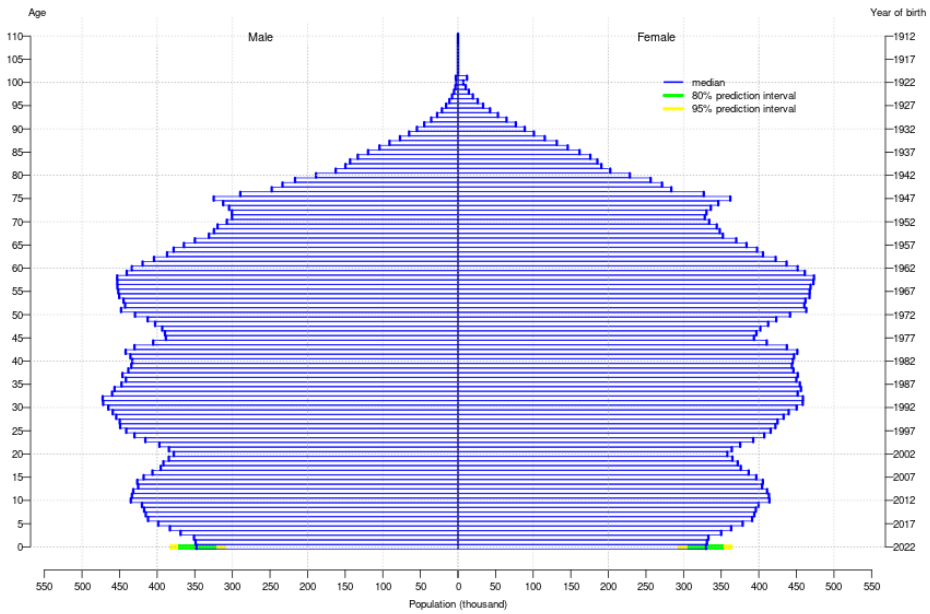


Figure 3: Population of the UK by age and sex in 2023

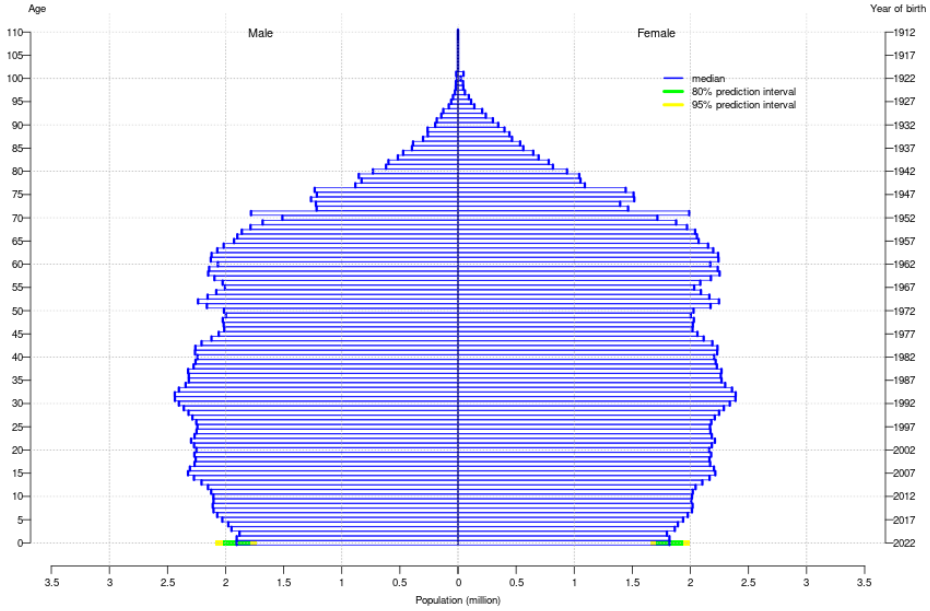


Figure 4: Population of the USA by age and sex in 2023

Commencing in the year 1912, the population of the UK has exhibited a predominance of females over males, a pattern that witnessed a notable escalation post-2017, thereby suggesting an extended life expectancy for females. A parallel trend was observed in the USA; however, in recent years, the gender ratio has approached equilibrium. These nations peaked in their respective populations in the years 1963 and 1991. Intriguingly, a decline in the male population was observed in both the UK and the USA in the year 1922, which might be the consequences of "World War I". Figure 5 represents the population pyramid of China.

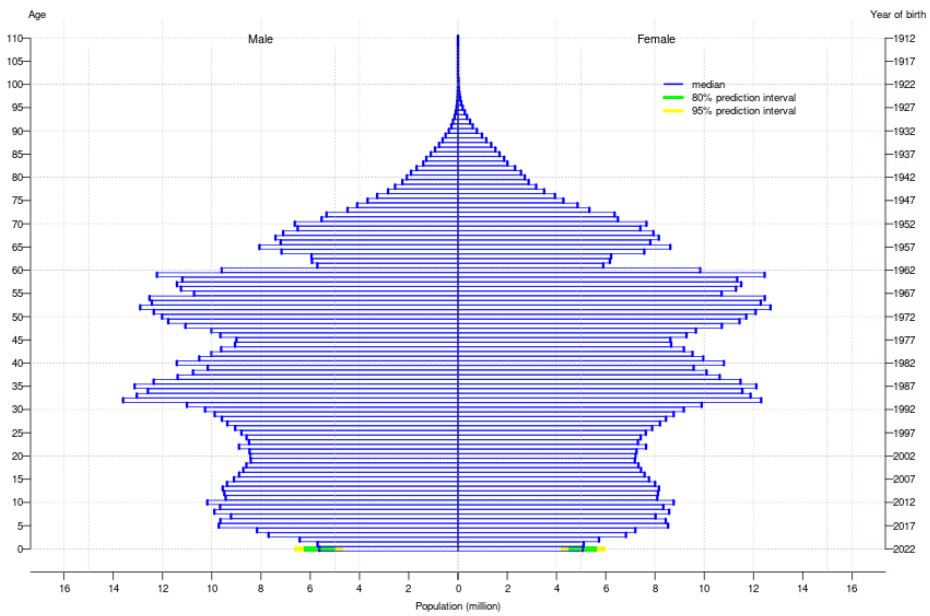
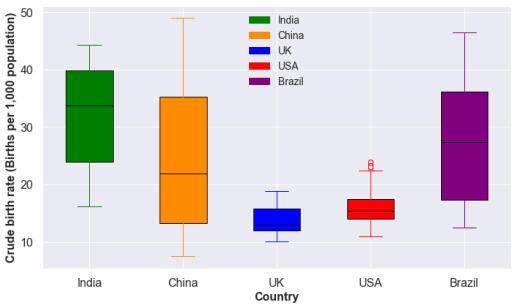


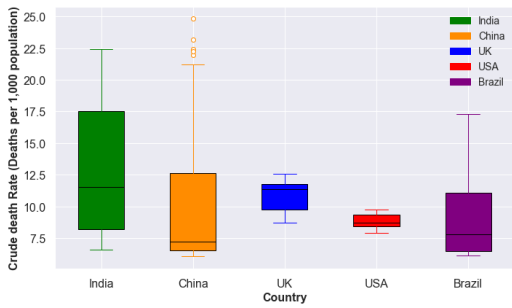
Figure 5: Population of China by age and sex in 2023

In contrast to the demographic trends observed in other nations, China exhibits a sustained majority of females, a phenomenon that has its origins in the early 20th century. Notably, after the year 1952, the population distributions of both genders have demonstrated a marked parallelism. An abrupt decline in the populations of both males and females between the years 2012 and 2022 suggests impending shifts within the societal framework. Such a decline may serve as an indicator of an increasingly aging population, a situation that could have been intensified by historical family planning regulations, such as the one-child policy, which was officially relaxed post-2016. On analyzing the population pyramids, it becomes evident that India and Brazil are characterized by youthful populations, while China leans toward an older age spectrum. The UK is undergoing an aging process, whereas the USA exhibits a heterogeneous age composition. Gender balance appears robust in India, Brazil, and the USA; however, notable gender disparities are present in China and the UK. Both the UK and the USA shows a similarity in male population for the year 1922, potentially attributable to the aftermath of World War I.Such demographic stability denotes the resilience exhibited by these nations amidst global fluctuations. India's population trajectory is ascendant, diverging from China's controlled decline. Influenced by policy and cultural factors, the demographic trends of major global entities such as China and India produce significant repercussions on a global scale.

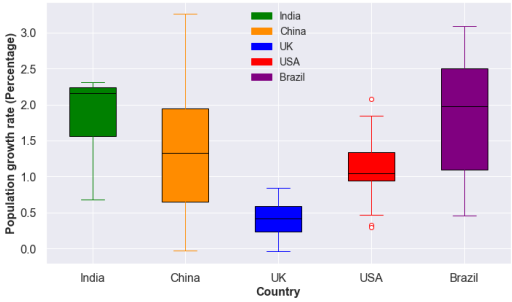
In the context of data analysis, the box-and-whisker plot or a box plot, serves as an efficient graphical representation to depict the distribution of data points across a specified metric. Various statistical metrics, notably the outliers, median, and primary data concentration within the box are visualized. Lines starting from the box depict the data's range, with white dots beyond these lines marking outliers. The box encapsulates approximately half of the data points and is segmented into five key statistical markers. The maximum value generally lies distant from the box (top whisker), while the minimum value is its counterpart on the opposite end (bottom whisker) [Tableau, 2023]. Following the minimum, the first quartile (Q1) accounts for data above 25%. The median, or second quartile, divides the data set. Lastly, the third quartile (Q3) represents data above the final 75% [Yi, 2021]. Figure 6 represents the box plots of CBR, CDR, population growth, life expectancy of both genders and median age population from 1950 to 2023. Figure 6 depicts that India leading in CBR, followed by Brazil and China. The UK and the USA, however, have lower CBR. Regarding CDR, India is the highest closely followed by the UK and the USA, suggesting significant death rates while China and Brazil report lower figures.



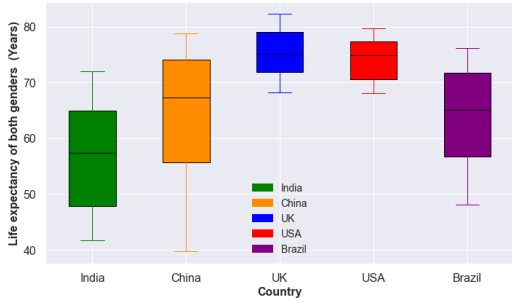
(a) Box plot of crude birth rate



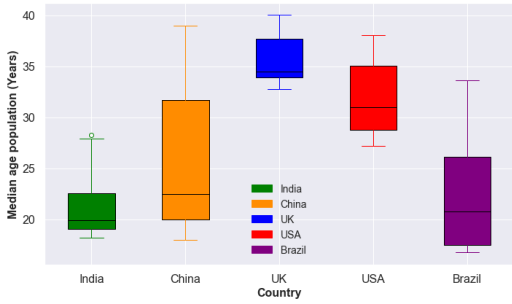
(b) Box plot of crude death rate



(c) Box plot of population growth



(d) Box plot of life expectancy of both genders



(e) Box plot of median age population

Figure 6: Population Growth based on World Population Prospects 2022

In terms of population growth, India and Brazil have high growth rates while the UK shows the lowest growth rate. China and the USA have stable growth rates as compared to other countries. Life expectancy, indicating healthcare quality and living standards, is higher in the UK and the USA. This contrasts with shorter life expectancies in China, Brazil, and especially India. Regarding median age population, the UK and the USA denotes older populations, influenced by lower CBR and longer life expectancy. On the other hand, India and Brazil shows younger median ages, influenced by higher CBR and migration. Therefore, the UK, the USA and China have lower birth rates, longer life expectancies, and older populations, while India and Brazil shows higher birth and death rates which denotes younger populations, and lower life expectancies. A detailed analysis of demographic variables such as CBR, Total Population Crude Net Migration are included in the following sessions.

2.2 Birth rates

In Figure 7 we present a bar chart of the total number of births in each decade for the five different nations under investigation.

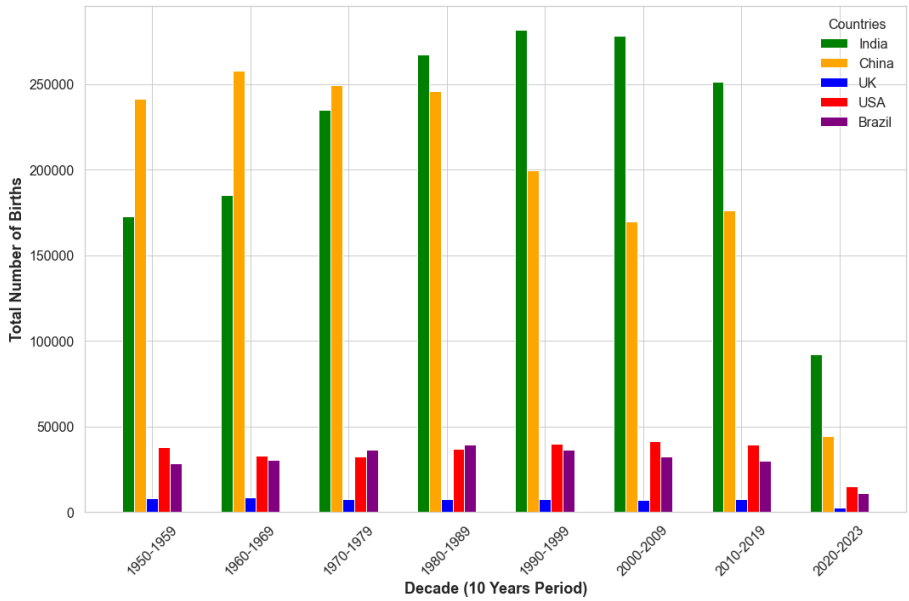


Figure 7: Analysis of birth trends over the decades

Although the fertility metrics for Brazil indicate a recent downtrend, it would be imprudent to disregard Brazil's capacity for future demographic expansion because each birth lays the groundwork for future growth. Concurrently, the UK represents demographic stability, exhibiting consistent birth rates from 1950 through to 2023. Such stability indicates a progressively aging population, thereby highlighting an unique demographic trajectory.

2.3 Total population

Figure 8 illustrate the evolution of the population of each country from 1950 to 2023.

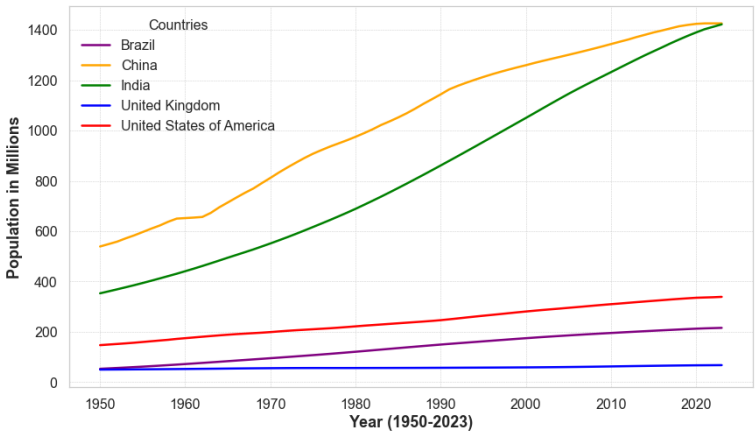


Figure 8: Comparison of total population offFive countries from the years 1950 to 2023

India, China, and Brazil are pivotal entities in the evolving global demographics. China led in birth rates from the period spanning 1950 to 1970; however, India outpaced China beginning in 1980 and has sustained its leadership. This development suggests that India may become the central hub for an expanding youthful demographic. The USA and Brazil have shown intermittent spikes in fertility rates. The USA was at the forefront until 1970, after which Brazil gained prominence starting in 1980. Throughout the years, every nation has witnessed a rise in its population. Initially, China appeared dominant in this aspect; however, the rapid population increase in India merits attention. The USA and Brazil, albeit prominent, lag behind these Asian giants. The growth observed in the UK remains relatively conservative, registering the lowest among these five countries. Annual population growth rates are not consistent and can be influenced by socio-economic and political factors. The consistent growth rate in India suggests the potential success of its population management strategies.

2.4 Crude Net Migration (CNMR)

Net Migration serves as a quantitative metric indicative of a nation's demographic stability, derived from the net difference between the numbers of individuals entering and leaving a country over a specified duration, and it is standardized for every 1,000 inhabitants of the country [UN.org]. The Crude Rate of Net Migration (CNMR) provides a more detailed perspective, incorporating not only the net migration but also any yearly modifications. This rate is similarly standardized per 1,000 inhabitants [Network, 2023]. This metric is pivotal in evaluating shifts in population dynamics.

CNMR values for China, the UK, and the USA are shown in Figure 9 (a) and for India and Brazil in Figure 9 (b). On examining global migration patterns, the USA and the UK emerge as leading centers for incoming populations. From 1950 to 2020, the USA's CNMR values varied between 1 and 7, with the 1990-2000 period showing relative stability, possibly due to the economic conditions during that time. The UK, however, experienced some CNMR variability but showed a rising trend after 1990, making it a popular migration destination. Brazil presents a neutral migration pattern, evidenced by its near-zero average and median values. In Asia, India and China display net outgoing population trends, as seen from their negative CNMR values. China's CNMR values remain stable, suggesting it consistently exports human capital. In contrast, India's CNMR values show more variation, hinting at potential future migration changes.

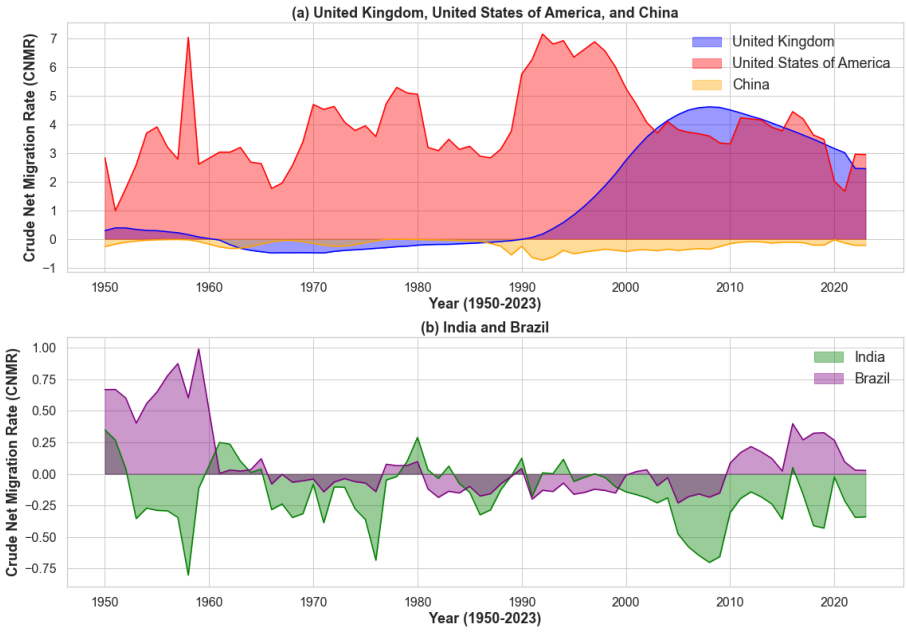


Figure 9: Analysis of Crude Net Migration Rate of five countries

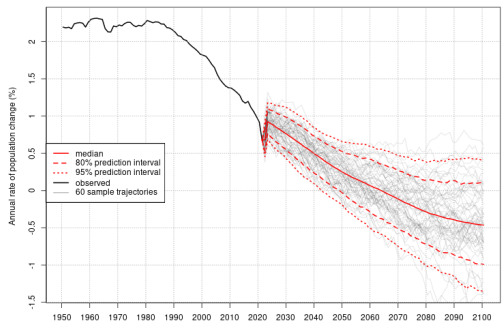
CNMR fluctuations might be driven by combinations of economic, social, or political reasons. Consequently, the USA and the UK attract global migration, while Brazil maintains balanced migration. Both China and India primarily export human capital. The USA's CNMR values suggest migration significantly influences its population. Since 2000, the UK's CNMR values have risen, indicating migration's major role in both the USA and the UK's population trends. For a detailed breakdown of CNMR values, refer to Appendix 2.

2.5 Population growth

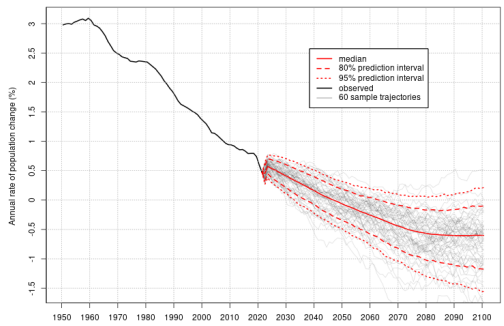
The average annual rate of change of population size during a specific period is the Population Growth Rate or per capita growth rate(R). It equals the difference between birth rate (B) and death rate (D) divided by the initial Population size (P) [LibreTexts, 2020].

$$R = \frac{B - D}{P}.$$

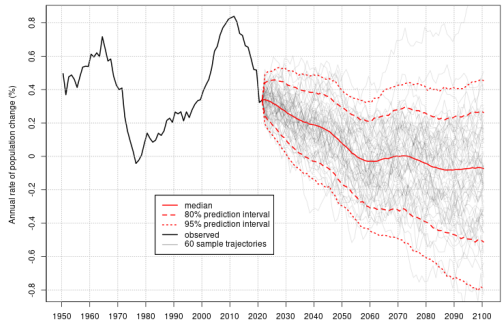
Figure 10 demonstrates the line charts of population growth rates of our chosen five countries, sourced from the website of United Nations Population Division [Nations, 2019]. The population growth of India show an early surge in the mid-20th century, declining to a more sustainable 0.5 percent rate, signaling a maturing demographic landscape, moving towards stability. Brazil transitioned from a high 3 percent growth rate to below 0.5 percent due to socio-economic changes and policy implementations aimed at population control.



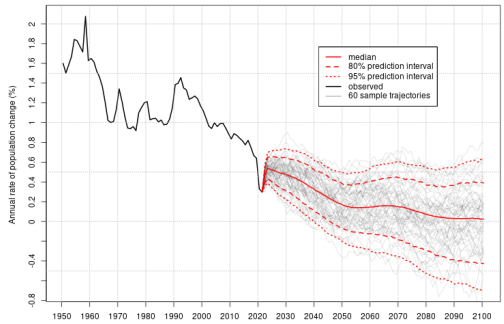
(a) Population Growth of India



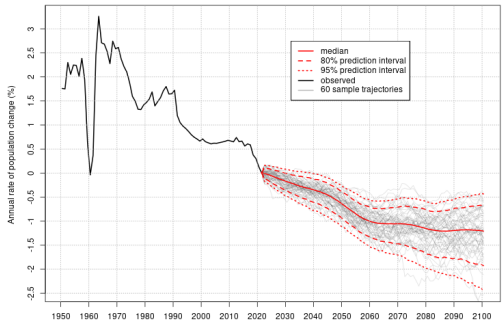
(b) Population Growth of Brazil



(c) Population Growth of the UK



(d) Population Growth of the USA

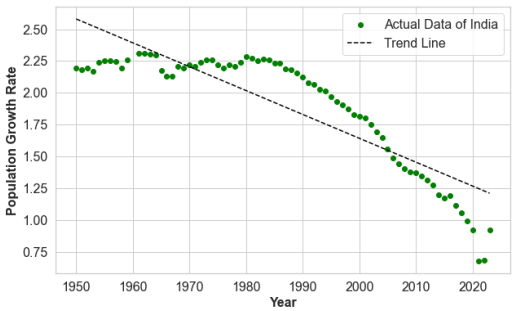


(e) Population Growth of China

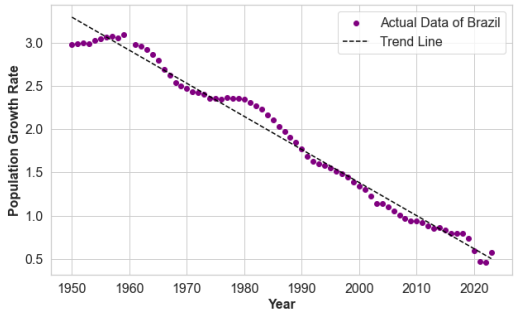
Figure 10: Population Growth based on World Population Prospects 2022[1]

The growth of China has fluctuated considerably, influenced by the implementation and modification of the one-child policy, showing peak in the 1970 and trough in 2020. The UK, though modest in population, exhibits a complex growth pattern characterized by periods of rapid growth and stagnation, necessitating detailed analysis. The USA has maintained steady growth between 1.5 percent and 0.2 percent, avoiding zero-growth and consolidating its stability in global demographics. Figure 11 shows long term trends in population growth across our selected countries.

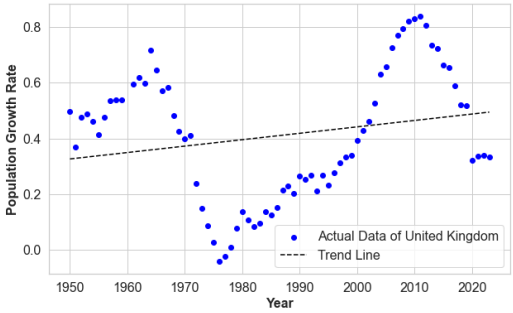
[1]Source: [Nations, 2019]



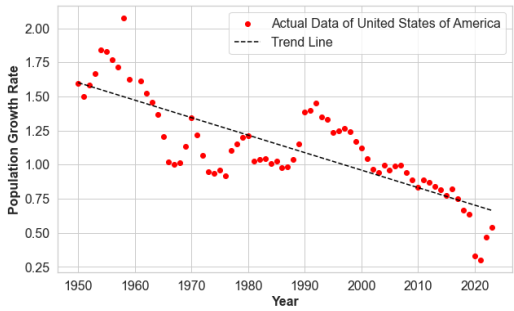
(a) Population Growth Trend of India



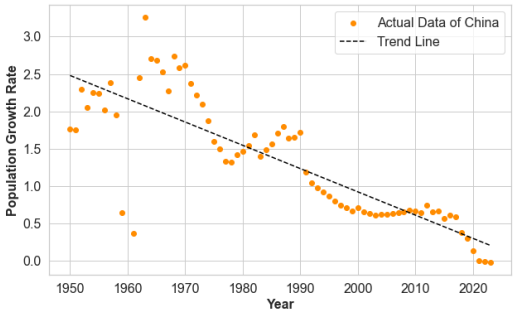
(b) Population Growth Trend of Brazil



(c) Population Growth Trend of the UK



(d) Population Growth Trend of the USA



(e) Population Growth Trend of China

Figure 11: Analysing Population Growth Trend using Linear Regression

The analysis reveals a long term declining population growth rate in India and a slow rising trend in the UK, as indicated by their respective trend lines. Similar downward trends are evident in the USA, Brazil, and China, the latter likely influenced by its population control measures. Fluctuations in population growth carry significant national implications. A reduced growth rate can ease resource strain but risks an aging population, affecting labour and social systems. Conversely, an increased rate can boost economic potential but may overburden public services and resources. Such demographic shifts are crucial for informed policy-making and sustainable growth. In Chapter 2, the data set is examined to derive preliminary statistical insights. Chapter 4 will implement this data set to forecast future population trends, based on the methodology described in Chapter 3.

Chapter 3 : Methodology

In this chapter, the focus is directed towards the exploration of machine learning techniques applied to the variables examined in Chapter 2, for predictive modeling. Machine Learning (ML) stands as a distinct sub-domain of Artificial Intelligence (AI). Unlike conventional problem-solving approaches, where established rules includes the linkage between queries and their appropriate responses, ML adopts a more explorative stance. Traditionally, these rules or patterns that generate answers from questions are presumed known. However, the ML framework diverges from this tradition.

In ML, these guiding patterns remain undisclosed and necessitate discovery. The ML methodology revolves around finding these patterns by analyzing the associations between queries and the correct answers, as provided during the training phase of the system. This distinction indicates the transformative nature of ML in the broader aspects of AI, highlighting its capability in addressing challenges through a more adaptive and self-improving mechanism, where the rules are not dictated but rather learned. Statistical Machine Learning (SML) offers a combination of both ML adaptability and computational ability with the robustness of statistical inference and modeling. This domain initiates statistical methodologies to cultivate models capable of learning from the data and subsequently make informed predictions or decisions. The "supervised," "unsupervised," and "semi-supervised" learning, rely on a solid statistical foundation [Xiong, 2023].

A distinguishing factor between SML and ML lies in their fundamental objectives. SML are designed to discover and explain connections between variables. The majority of the work of the SML includes creating models of the $P(x)$, the probability distribution and producing outcomes based on inferences made using such a model. ML models, on the other hand, are designed to provide exact predictions without the need for explicit programming [turintech, 2023]. Such contrasts highlight the significance of both disciplines, particularly for analysts seeking complete data-driven insights. The ML techniques that have been implemented in this dissertation includes linear regression, ARIMA, exponential smoothing, prophet model, and the Gradient Boosting Regressor. Each of these methodologies will be described in detail to ensure an extensive understanding of their applicability and efficacy.

3.1 Linear regression

'Regression is a method for extracting a pattern from noise by attempting to predict the value of a certain number of continuous desired variables t from the value of a D-dimensional vector x of input variables. In their most basic form, linear regression models are also linear functions of the input variables. However, a far more useful class of functions may be formed by choosing linear combinations of a specified number of nonlinear functions of the input variables (basis functions). These models are linear functions of the parameters and have simple analytical features, but they can also be nonlinear relative to the input variables. [Jordan et al., 2006]'. 'For a data set, $\mathcal{D} = (x_n, y_n)_{n=1}^{M}$, The linear regression can be expressed in the form of, $y = h(x) + \epsilon$ where $x_n \in \mathbb{R}^d$ is the independent variables or features. $y_n \in \mathbb{R}^s$ is the dependent variable and $\epsilon \in \mathbb{R}^s$ is the noise. The average noise is assumed to be zero and if not zero, it may absorbed into h. Similarly, consider a data set consisting of U observations of x_u, where $u = 1, ..., U$, along with the corresponding target values t_u, the objective is to predict the value of t for a new value of x. In general, this may be accomplished by immediately building an appropriate function $y(x_u)$, the values for new inputs x_u represent predictions for the associated values of t_u. The purpose, from a probabilistic perspective, is to model the predictive distribution $p(t_u|x_u)$, which reflects the uncertainty about the value of t for each value of x_u. Using this conditional distribution, predictions of t_u for every new value of x_u could be produced in order to minimize the expected value of a suitable loss function. On broadening the class of models by taking into account linear combinations of constant nonlinear functions of the input variables. The total number of components in this model will be 'M', if the maximum value of the index 'j' is represented by $M - 1$. The component w_0 allows for any constant bias in the data and can be referred to as a bias parameter. It could be advantageous to establish an extra fake 'base function' $\phi_0(x) = 1$ such that,

$$y(x, w) = \sum_{j=1}^{M-1} w_j \phi_j(x) = w^T \phi(x)$$

where $w = (w_0, ..., w_{M-1})^T$ and $\phi = (\phi_0, ..., \phi_{M-1})^T$ and $\phi_j(x)$ are the basis functions. Many practical pattern recognition programs conduct some rudimentary pre-processing, or feature extraction, on the unprocessed variables. The features may be defined using the basis functions if the original variables are represented by the vector x. Consider a data set with inputs $X = x_1, ..., x_N$ and target values, $t_1, ..., t_N$. We aggregate the target variables t_n into a column

vector indicated by t. Then the sum-of-squares error function can be represented as,

$$E_D(w) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - w^T\phi(x_n)\right)^2.$$

By minimizing $E_D(w)$ we find that for x^k elements of the basis set

$$w = A^{-1}z$$

where

$$z_i = \frac{1}{M}\sum_{n=1}^{M} y_n x_n^i$$

$$A_{i,k} = \frac{1}{M}\sum_{n=1}^{M} x_n^{k+i}$$

'[Jordan et al., 2006]

3.2 Auto Regressive Integrated Moving Average (ARIMA) Model

A time series data is a sequence of data points that occur in successive order over some period of time [Influxdata, 2022]. The ARIMA model is a forecasting approach of time series data. The mean and variance of a stationary time series data are unaffected by the time at which it is examined. A trend occurs when there is a long-term rise or reduction in data, and it does not have to be consecutive. A time series is adversely impacted by seasonal elements such as the time of year or the day of the week, (a seasonal pattern is formed). Seasonality has a definite and predictable periodicity while a cycle occurs when the data shows increases and falls with a variable frequency. The time series with trends or seasonality are not stationary. Many of the computations cannot be made and give the desired results if a trend arises and stationarity is not satisfied. Hence the first step is to check the stationarity of the data and if it is not stationary it could be transformed into a stationary data using methods such as differencing or log transformation. Differencing can be computed as the differences between consecutive observations and it can help in the stabilization of a time series mean by removing variations in the level of the time series, hence eliminating (or lowering) trend and seasonality. When working with time series lags, the backward shift operator B can be written as, $y_{t-1} = By_t$. The differencing methods includes first-order differencing, second-order differencing and seasonal differencing [Hyndman and Athanasopoulos, 2016]. The first differencing can be written as,

$$y_t^{'} = y_t - y_{t-1} = y_t - By_t = (1 - B)y_t.$$

Similarly for second-order differencing, $y_t^{''} = (1-B)^2 y_t$. The Autocorrelation Function (ACF) can also be used to detect non-stationary time series. The ACF of a stationary time series approaches zero relatively rapidly, but the ACF of non-stationary data declines slowly. There are many types of time series analysis used for forecasting, the prominent models used are SARIMA, SARIMAX, ARIMA and exponential smoothing where the first three are defined for the data sets with high seasonality and trends and the latter fits for data sets without seasonality. The components of ARIMA serve as parameters with standard notation. A conventional notation for ARIMA models would be ARIMA with *p, d,* and *q*, where p is the number of lag observations in the model, commonly known as the lag order, d is the number of times the raw observations are differenced; also known as the degree of differencing. While q denotes the size of the moving average window, also known as the order of the moving average. In a multiple regression model the desired variables are forecasted using a linear combination of variables while in an autoregression model the desired variables are forecasted using a linear combination of the variable's historical values. A moving average model, rather than utilizing prior predicted values in a regression, employs past forecast errors in a regression-like model. A non-seasonal ARIMA model is obtained by combining differencing with autoregression and a moving average model.

$$y_t^{'} = c + \phi_1 y_{t-1}^{'} \cdots + \phi_p y_{t-p}^{'} + \theta_1 \epsilon_{t-1} \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

where $y_t^{'}$ is the differenced series, ϕ_p is the autoregressive terms, y_t is lagged values, θ_q is the coefficient of moving average and ϵ_t are lagged errors. This is known as an ARIMA(p, d, q)model. AutoRegression(p) is represented by

$$(1 - \phi_1 B - \cdots \phi_p B^p).$$

The dth order differencing can be represented as,

$$(1-B)^d y_t.$$

Moving average(q) is denoted by the equation below,

$$c + (1 + \theta_1 B + \cdots + \theta_q B^q)\epsilon_t.$$

The maximum likelihood of ARIMA model for T number of observations is similar to the least squares estimates obtained by minimising the errors,

$$\sum_{t=1}^{T} \epsilon_t^2.$$

The p,d and q parameters can be determined using Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). AIC and BIC can be calculated from the equations below,

$$AIC = -2\log(L) + 2(p + q + k + 1.$$

$$BIC = AIC + \frac{2(p + q + k + 1)(p + q + k + 2)}{T - p - q - k - 2}.$$

where L is the likelihood of the model, k is an intercept depending upon c. $k = 1$, if $c \neq 0$ and $k = 0$, if $c = 0$. [Hyndman and Athanasopoulos, 2016]

3.3 Gradient Boosting Regressor (GBR) Model

The basic idea behind boosting techniques is that we build a model on the training data set first, then build a second model to correct the faults in the first model. The GBR model is based on the concept of boosting in data science, in which a model is constructed by attempting to repair errors produced by earlier models. It utilizes the principles of Gradient Descent to minimize mistakes and attempts to transform weak learners into strong learners. GBR begins by creating a single leaf, rather than a tree, to reflect an initial approximation of predictions for all data. GBR constructs trees depending on the preceding tree's mistakes (residuals) and continue to build trees until it has built the amount of trees requested or until the extra trees fail to be a better match than the original ones. The objective is to find a predicted value whose loss function is minimum. The loss function of GBR with observed value y_i and predicted value γ_i can be represented as,

$$L = \frac{1}{n}\sum_{i=0}^{n}(y_i - \gamma_i)^2.$$

The initial stage in GBR is to create a base model that predicts the training values of the data set. For simplicity, the average of the target column is calculated and assumed as the forecasted number. The pseudo residuals (observed value - predicted value), are then calculated. In the following stage, a model based on these pseudo residuals are created and made predictions in order to decrease these residuals. Doing so will eventually increase our accuracy of the model and prediction capacity. As a result, we will produce new predictions using the residual as the

target. We determine the output values for each leaf of our decision tree in this stage which means there is a chance that one leaf will get more than one residual, thus we must determine the average of all the leaves. This is the final phase in which we must update the prior model's predictions using the equation below,

$$F_m(x) = F_{m-1}(x) + \nu_m h_m(x),$$

where m is the number of decision tress made, $F_m(x)$ is the new prediction, ν_m is the learning rate and $h_m(x)$ is the tree made on residuals Saini [2021].

3.4 Prophet Model

"Prophet is a procedure for forecasting time series data based on an additive model where non-linear trends are fit with yearly, weekly, and daily seasonality, plus holiday effects".[Robson, 2020] 'This is a sophisticated forecasting algorithm developed by Facebook to anticipate daily data with weekly and annual seasonality, as well as holiday impacts. It was later enhanced to accommodate other types of seasonal data. It performs well with time series with high seasonality. It attempts to fit a variety of linear and nonlinear temporal functions as components. The Holt-Winters technique employs the similar concept of modeling seasonality as an additive component. In effect, we are framing the forecasting problem as a curve-fitting exercise rather than explicitly studying the time-based dependency of each observation inside a time series. Prophet is a nonlinear regression model in the form,

$$y_t = g(t) + s(t) + h(t) + \epsilon_t$$

where $g(t)$ represents a piecewise-linear trend (or "growth term"), $s(t)$ represents the different seasonal patterns, $h(t)$ represents the holiday impacts, and ϵ_t is a white noise error term. In this method, at certain times, the growth rate is permitted to fluctuate. A logistic function can be used to establish an upper constraint on the trend if desired. The seasonal component is made up of Fourier terms from the relevant time periods. Order 10 is used for annual seasonality by default, whereas order 3 is used for weekly seasonality. As simple dummy variables, holiday effects are included' [Hyndman and Athanasopoulos, 2016].

3.5 Exponential Smoothing

'Simple exponential smoothing assumes that the time series does not fluctuate about the mean value or level. As a result, it cannot be used with series that contain trends, seasonality,

or both. Holt's exponential smoothing is one step above ordinary exponential smoothing in that it allows time series data to include a trend component but it is still incapable of dealing with seasonal data'[Ariton, 2021]. Holt-Winter's exponential smoothing extends Holt's exponential smoothing by allowing for the incorporation of seasonality. This approach has two variants that vary in the context of the seasonal component. When the seasonal variations are essentially constant throughout the series, the additive technique is favoured, but the multiplicative method is preferable when the seasonal variations change according to the level of the series. The seasonal component is represented in absolute terms in the scale of the observed series using the additive approach, and the series is seasonally adjusted in the level equation by removing the seasonal component. Within each year, the seasonal component will be close to zero. The seasonal component is stated in relative terms (percentages) in the multiplicative technique, and the series is seasonally adjusted by dividing through by the seasonal component. The Holt-Winters seasonal technique consists of the forecast equation and three smoothing equations such as one for the level l_t, one for the trend b_t, and one for the seasonal component s_t, each with associated smoothing parameters. The additive method can be represented as,

$$\hat{y}_{t+h|t} = l_t + hb_t + s_{t+h-m(k+1)},$$

$$l_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(l_{t-1} + b_{t-1}),$$

$$b_t = \beta^*(l_t - l_{t-1}) + (1-\beta^*)b_{t-1},$$

$$s_t = \gamma(y_t - l_{t-1} - b_{t-1}) + (1-\gamma)s_{t-m}$$

where k is the integer part of $\frac{h-1}{m}$, h is the number of periods for which forecasting is performed, m represents number of periods in a season, α is the smoothing parameters of the level component, β and β^* are the smoothing parameters of the trend component, γ and γ^* are the smoothing parameter of the seasonal component. This guarantees that the estimates of the seasonal indices used for forecasting are from the final year of the sample. For time t, the level equation displays a weighted average of the seasonally adjusted observation $(y_t - s_{t-m})$ and the non-seasonal forecast $(l_{t-1} + b_{t-1})$. Holt's linear technique is similar to the trend equation. The seasonal equation depicts a weighted average of the present seasonal index $(y_t - l_{t-1} - b_{t-1})$ and the seasonal index of the same season last year (m time periods ago) [Hyndman and Athanasopoulos, 2016]. The equation for seasonal component can be denotes as,

$$s_t = \gamma^*(y_t - l_t) + (1-\gamma^*)s_{t-m}.$$

3.6 Evaluations Metrics used in models

Evaluation metrics are quantitative measures used to analyze a statistical or machine learning model's performance and efficacy. These metrics give information about how well the model performs and serve in the comparison of different models or algorithms. It is critical to examine a machine learning model's prediction capacity, generalization capabilities, and overall quality while assessing it. These features are measured using the evaluation metrics [Srivastava, 2019]. When forecasting the outcome of a particular event, the coefficient of determination (R^2) is a statistical measurement that assesses how variations in one variable may be explained by changes in a second variable. In other words, it quantifies the strength of the linear relationship between two variables and is widely depended on by analysts when doing trend research. The equation for calculating coefficient of determination is given below,

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

where n is the number of observations, $\hat{y}$ is the predicted value of the model, y_i is the ith observed value and $\bar{y}_i$) is the mean value. The numerator is the sum of the squared difference between the actual observed value and the mean value. When all observations are added together, the total variance in the data is obtained whereas the denominator is the sum of the squared difference between the actual observed value and its projected value. When the residual sum of squares is calculated across all data, it gives the unexplained variance by the model. The coefficient is represented by a value upto 1.0 (100%). Thus, greater the value of R^2, the better the model's prediction and strength [Nevil, 2021]. The Mean Absolute Error (MAE) is the average of the differences between the original (y_j) and predicted values ($\hat{y_j}$). It indicates how much the predictions were divergent from the actual output. However, they provide no indication of the direction of the inaccuracy, i.e. whether we are underestimating or overestimating the data. It is denoted mathematically as,

$$MAE = \frac{1}{N}\sum_{j=1}^{N}|y_j - \hat{y}_j|.$$

Mean Squared Error (MSE) is very similar to MAE, with the only distinction being that MSE takes the average of the square of the difference between the actual and forecasted values as represented in the following equation,

$$MSE = \frac{1}{N}\sum_{j=1}^{N}(y_j - \hat{y}_j)^2.$$

MSE has the benefit of being easier to compute the gradient, whereas Mean Absolute Error needs complex linear programming methods to compute the gradient. As we square the error, the influence of greater errors becomes more noticeable than smaller ones, therefore the model may now focus more on the larger errors [Mishra, 2018].

The outcomes derived from applying these methodologies to the selected countries during the specified time interval have been evaluated based on specific performance metrics. Detailed insights into these outcomes, including potential discrepancies and errors, are clearly elaborated upon in Chapter 4, followed by conclusions drawn in Chapter 5.

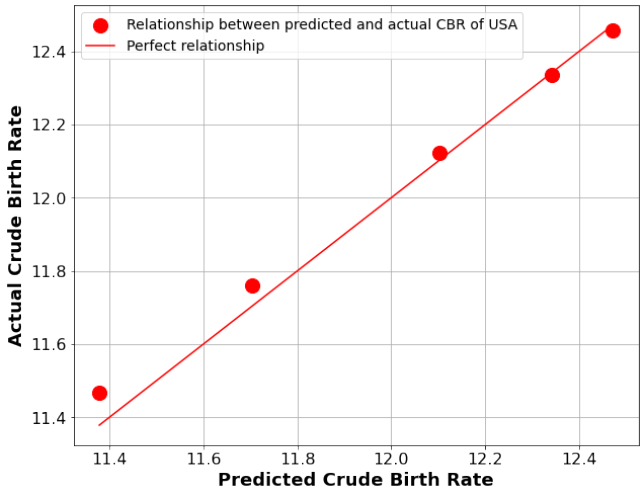
Chapter 4 : Theory and Results

In this chapter, the dissertation presents results derived from the variables identified from the literature review. Utilizing the data set elaborated in Chapter 2, these variables are methodically examined through the analytical frameworks as described in Chapter 3. The objective was to forecast demographic metrics like CBR, CDR and total population for the five chosen countries. A variety of methods, including linear regression, time series analysis, random forest, and exponential smoothing, were utilized. For CBR, techniques such as linear regression, ARIMA, and the GBR model were applied. The GBR model was exclusively used for CDR, while linear regression was chosen for total population predictions. The Prophet model was employed to forecast populations. All models, except the Prophet, were trained on subsets of data and then tested on subsequent sets. Their accuracy was gauged using statistical metrics, and visual tools facilitated the comparison of predicted and actual values. Specifically, in the time series analysis, the ARIMA model was optimized for each country using the AIC values. For all the models used, the accuracy is assessed using prominent performance metrics such as the R^2, MSE, and MAE, by comparing predicted values against actual data. It is pertinent to note that greater emphasis has been placed on total population and CBR as compared to CDR. This prioritization is due to the inherent significance of birth rates in contributing to further population increments as a result of reproduction.

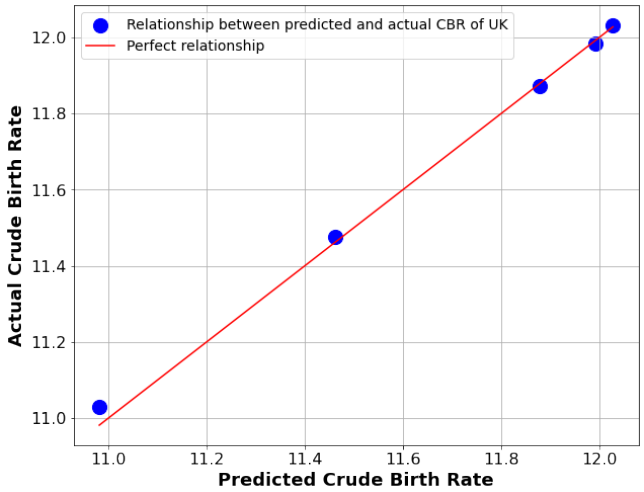
4.1 Crude Birth Rate

The linear regression model have been employed on the data set for predicting CBR using the training data from 2003 to 2013 and testing data from 2014 to 2018. Figure 12 illustrates the predicted CBR against actual CBR of five countries over a five year period from 2014 to 2018. This time period is chosen for prediction so as to evaluate the time-agnostic capability of the model. Each data point on the graph signifies one of these years in a decreasing manner, and a red line represents the perfect correlation between the forecasted and actual demographic figures. During the examined time-frame, a downward trajectory in the CBR is evident across all the countries under study. Nonetheless, the trajectories exhibit variances. China and Brazil shows deviations from a linear pattern, which could be attributed to demographic control measures or policy enactments. In contrast, the USA and India display consistent downward trends in CBR without any deviation from the linear pattern. A noteworthy incident is observed in Brazil, where the CBR decreased abruptly from 14.7 to 14.2 and sustained a reduction. Validation procedures such as R^2, MAE and MSE verified the ability of the model to reliably estimate CBR for the time period. Table 1(a) shows the values of the performance metrics in

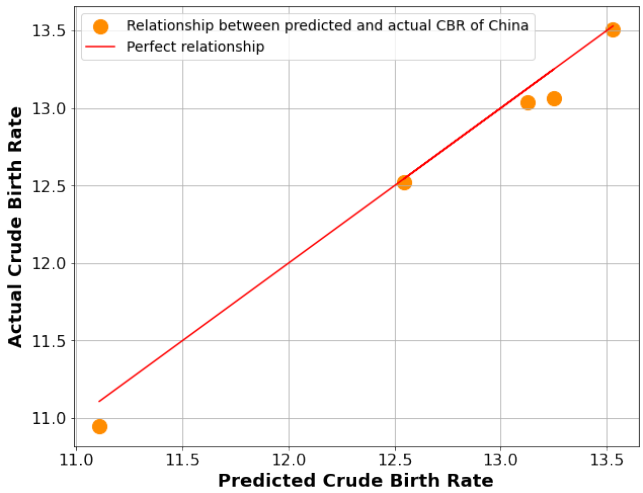
predicting CBR using linear regression.



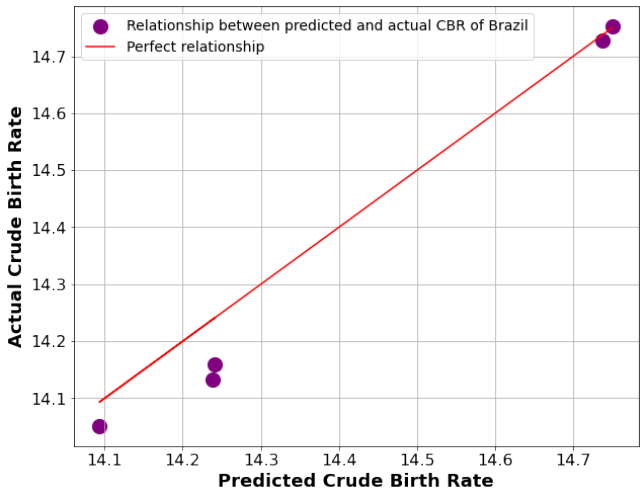
(a) Crude Birth Rate of the USA



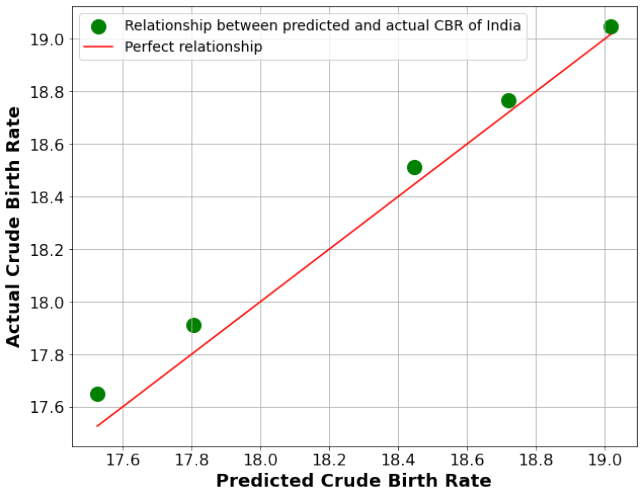
(b) Crude Birth Rate of the UK



(c) Crude Birth Rate of China



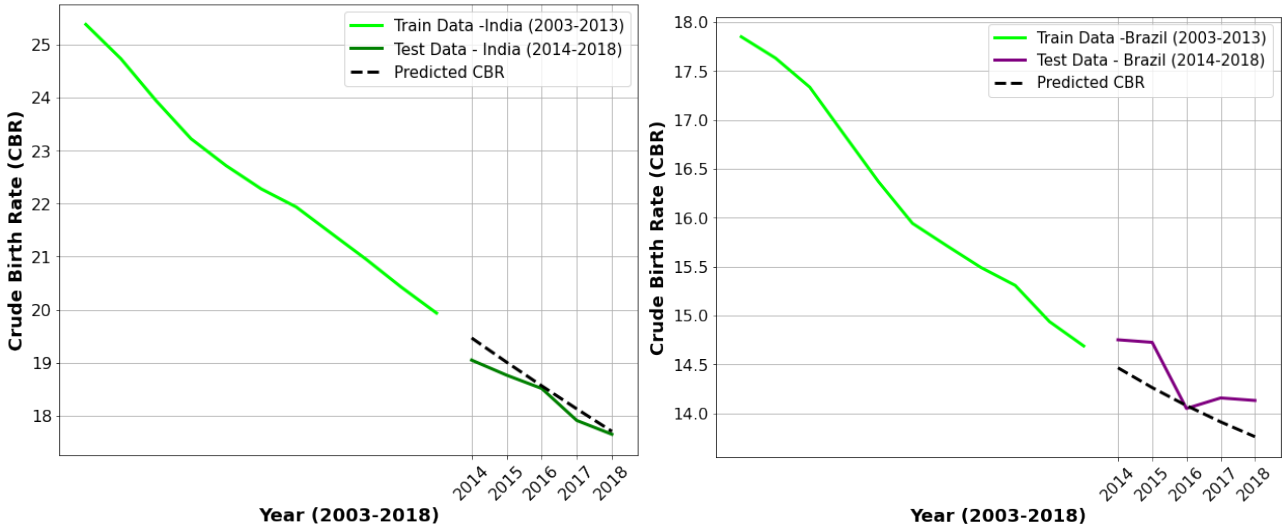
(d) Crude Birth Rate of Brazil



(e) Crude Birth Rate of India

Figure 12: A plot of the actual CBR against the predicted CBR using linear regression for each of the countries under consideration from 2014-2018.

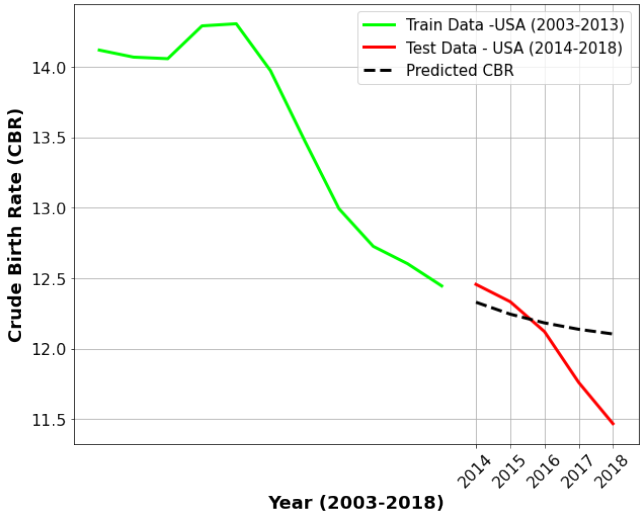
In the empirical analysis based on these metrics, the model demonstrates robust performance across the selected nations, particularly with a high degree of accuracy observed for the UK. For both China and the USA, the R^2 values endorse a satisfactory model fit. In the cases of Brazil and India, there exists a fine similarity between the predicted and observed values. Additionally, the model decently explains the variance in CBR, certifying a dependable fit and its possibility for extrapolation to other time periods, either retrospective or prospective, without reducing predictive accuracy. Getting into the next phase of analysis, time series ARIMA modeling was executed on the dataset for predicting CBR for the same time period. Figure 13 shows the ARIMA model implemented to predict CBR from 2014 to 2018 for the quintent of nations.



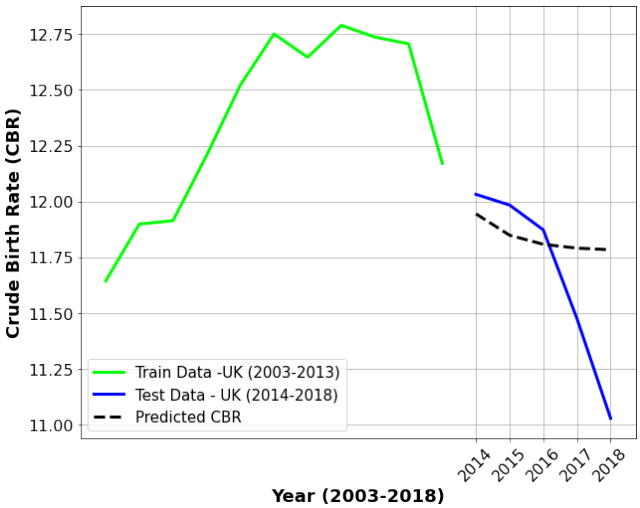
(a) Crude Birth Rate of India

(b) Crude Birth Rate of Brazil

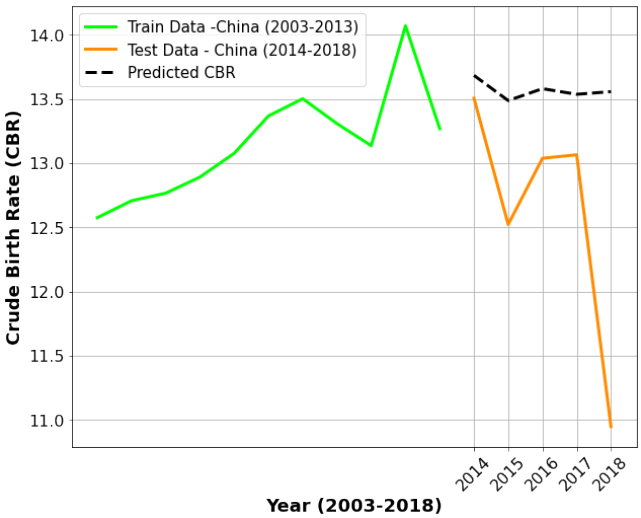
This particular model was selected due to the absence of seasonality in the data set under scrutiny. Despite efforts to transform the data into a stationary data set and optimize the ARIMA parameters *(p, d, q)* through criteria such as AIC and BIC, the model resulted in an average R^2 value of -0.033. Table 1(b) shows the values of the performance metrics in predicting CBR using the ARIMA model. For India, the model exhibited a commendable level of fit, as indicated by a higher R^2 value and a relatively low MSE and MAE. On the other hand, the model's performance is not up to par for the UK and the USA, resulting in lower R^2 values. In addition to that, China and Brazil shows sub-optimal performance of the model and negative R^2 values were observed in their case, suggesting that the model might not be well-suited for forecasting the CBR in countries with an unexpected fluctuation in their values. The R^2 values observed across the selected countries shows variable effectiveness of the model. In spite of the application of hyper parameter tuning and the incorporation of auto correlation to enhance parameter selection, no significant improvement in predictive accuracy was noted.



(c) Crude Birth Rate of the USA



(d) Crude Birth Rate of the UK



(e) Crude Birth Rate of China

Figure 13: A plot of the actual CBR along with the predicted CBR using the ARIMA model for each of the countries under consideration from 2014-2018.

In the study conducted by Otoom [2021], a comparison of 17 machine learning models for population prediction was undertaken, and the Random Forest model emerged as the most efficacious. Following this methodology, an attempt was made in this research to predict the CBR using the Random Forest model but the model resulted in negative R^2 values for all five countries, indicating the inadequacy or data anomaly of the model. A data anomaly, also known as an "outlier", refers to a data point or series of data points that deviates significantly from the standard observations, rendering it in contradiction with the expected pattern or trend of the larger data set [Soloveichik, 2023].

Moving onto the GBR model to predict the CBR of our five selected countries from 2008 to 2018. We have trained the data from 1998 to 2007 and tested it with the data from 2008 to 2018.

The choice of this ten year period of is due to two reasons. Firstly, the GBR model determines the dependency of other features of the data set excluding CBR and CDR. Secondly, a transition in migration and population control policies occured in this period. Figure 14 depicts the prediction of CBR using the GBR model for the same time period. The algorithm utilized all available features within the data set, excluding CBR and CDR, to generate predictions for the CBR. This approach facilitated a better understanding of the relationships between the selected features and CBR.

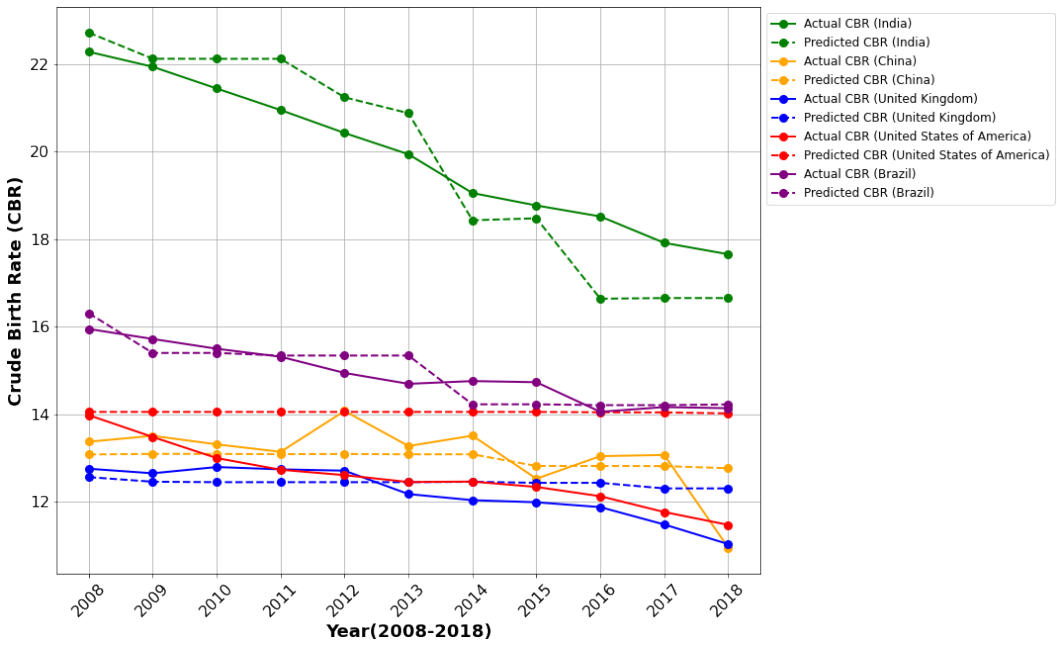


Figure 14: A plot of the actual CBR (training data from 1998-2007, testing data from 2018-2018) along with the predicted CBR using the Gradient Boosting Regressor model for each of the countries under consideration from 2014-2018.

The predictive performance of the model is validated through the established metrics. Table 1(c) shows the values of the performance metrics in predicting CBR using the GBR model. The research indicates variable model performances across nations. In India, the model significantly captures CBR variations with an R^2 value of about 58.5%, meaning the model represents a substantial amount of the observed variance in the actual CBR. This is further supported by the low MAE and MSE, highlighting the accuracy of the model as compared to actual data. The model's results for China and the UK are satisfactory, with R^2 values around 30%, suggesting a fair representation of the observed variance, further evidenced by low MAE and MSE values. However, for the USA, the results are less consistent. An R^2 value of roughly -2.63 indicates the limitations of the model, especially post-2015, reflected by high MAE and MSE values. This suggests that features other than CBR and CDR might not significantly affect the CBR of the USA from 2008 to 2018. In comparison, the R^2 value of Brazil is 52.2%, confirming the ability of the model to capture more than half of the variance in CBR, similar to India.

In summary, the model is precise for countries like India and Brazil, as shown by their high R^2 values and further supported by low MAE and MSE. China and the UK show moderate R^2 values, suggesting a 30% reliance of the features of the data set on CBR prediction. In contrast, the analysis for the USA indicates less influence of these features for the given period. For all the five countries the CBR is decreasing yearly in all the models implemented over a

Table 1: Comparison of the performance metrics of different models implemented for predicting CBR for the countries under consideration.

Errors in Crude Birth Rate (CBR)									
Countries	(a) Linear Regression			(b) ARIMA			(c) GBR		
	MSE	MAE	R^2	MSE	MAE	R^2	MSE	MAE	R^2
Brazil	0.004	0.048	0.959	0.099	0.279	-0.037	0.184	0.360	0.522
China	0.014	0.096	0.983	1.658	0.953	-1.091	0.401	0.473	0.307
India	0.007	0.074	0.975	0.057	0.197	0.790	0.997	0.889	0.585
UK	0.001	0.016	0.996	0.140	0.271	0.027	0.211	0.406	0.331
U.S.A	0.002	0.037	0.982	0.115	0.258	0.143	1.699	1.169	-2.628

period from 2003 to 2018. Also, the linear regression model for the CBR prediction could be implemented in CDR for presenting the future.

4.2 Crude Death Rate

Figure 15 depicts the predicted CBR using the GBR for the countries under consideration over a period of five years from 2013-2018.

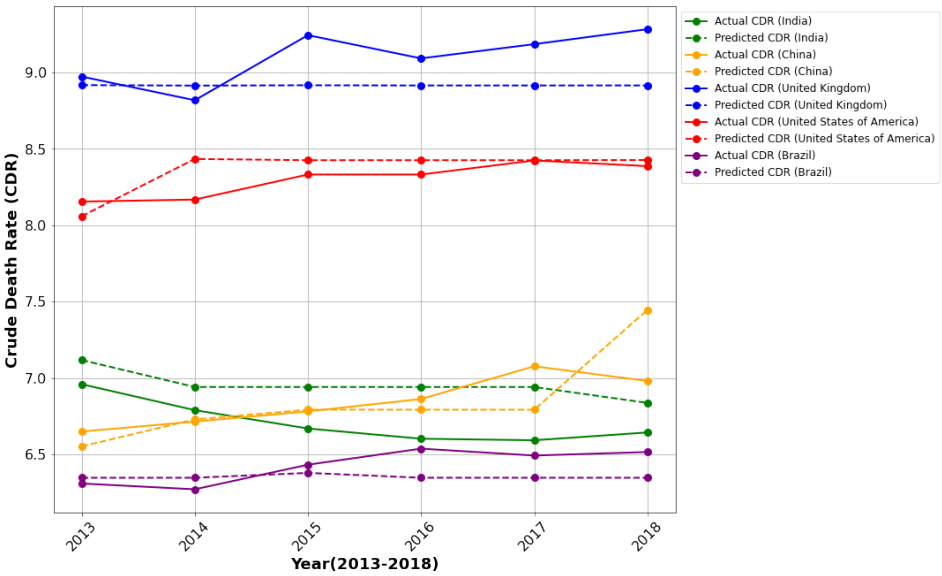


Figure 15: A plot of the actual CDR (training data from 2008-2012, testing data from 2013-2018) along with the predicted CDR using the Gradient Boosting Regressor model for all five countries from 2013-2018.

For analysing CDR we have implemented the GBR model. The training data rum from

2008 to 2012 and testing data runs from 2013 to 2018. The CDR for these five countries exhibited varied trends from 2013 to 2018. While the differences between the predicted and actual values were minimal, indicating that the predictions of the model closely matched the observed data. Table 2 shows the values of the performance metrics in predicting CDR using the GBR model for the time period from 2013 to 2018. Our analysis using the gradient boosting regressor model yielded decent results for predicting the CDR across the five countries.

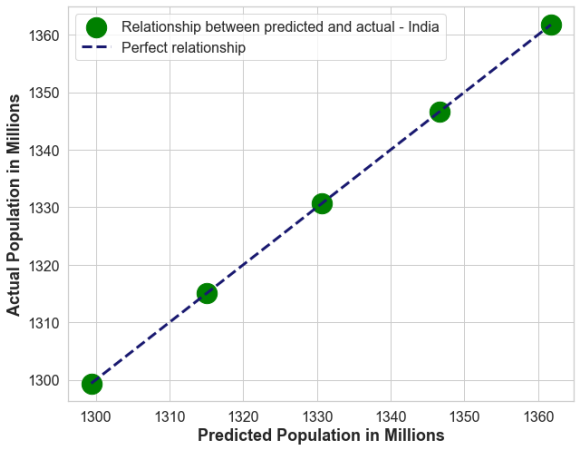
Table 2: Performance metrics implemented for predicting the CDR for the countries under consideration from 2013-2018 using the Gradient Boosting Regressor model.

Errors in Crude Death rate			
Countries	MSE	MAE	R^2
Brazil	0.016	0.112	-0.536
China	0.052	0.157	-1.366
India	0.066	0.244	-2.954
United Kingdom	0.060	0.216	-1.289
United States of America	0.016	0.098	-0.560

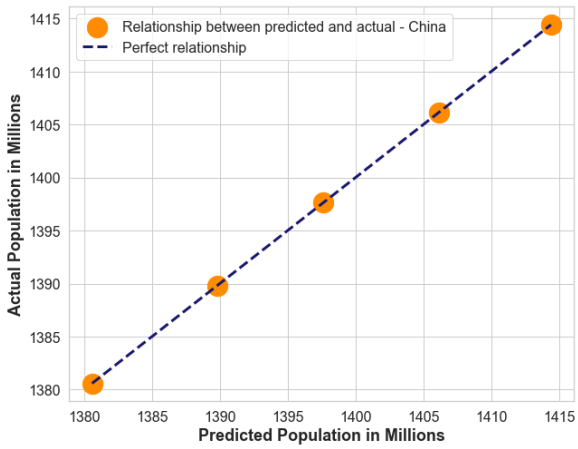
In an examination of the CDR performance metrics in table 2, the low MAE and MSE values of India suggest a satisfactory prediction despite its negative R^2 values. Apart from the negative R^2 values of all countries they all hold low MAE and MSE values. Transitioning to the GBR analysis, the USA's post-2015 predictions closely align with the actual values, indicating a good data pattern recognition. The UK's model, post-2015, deviates further from actuals. In conclusion, for almost all countries CDR is not influenced by any other features in the data set from 2013 to 2018. The model achieved an average R^2 value of -1.341 over all the countries, confirming its effort in capturing the correlation between CDR and the features. Each variable within the data set only has a small influence on the CDR for the quintet of nations. Overall, while each country showed unique fluctuations, a common pattern was the upward trend or stabilization of the CDR in the latter part of the period under study.

4.3 Total Population

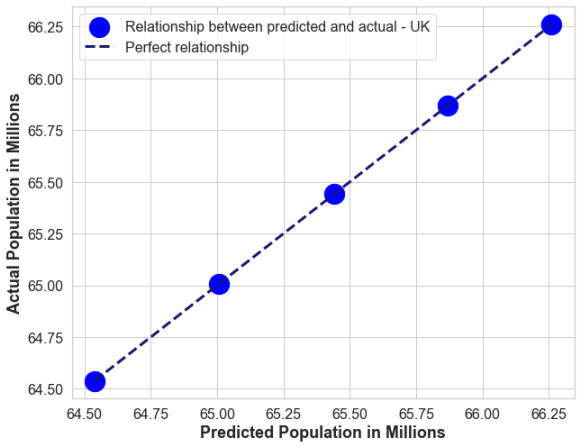
Similarly, the linear regression model is initially employed for predicting the total population of five countries over a five year period from 2014 to 2018. Figure 16 illustrates the predicted total population against the actual total population of five countries from 2014 to 2018. In Figure 16, the five dots represents five years from 2014-2018 and the dotted line indicates the perfect relationship between the predicted and actual population values. The model demonstrated an outstanding precision in predicting the total population within the scrutinized temporal scope for the quintet of nations.



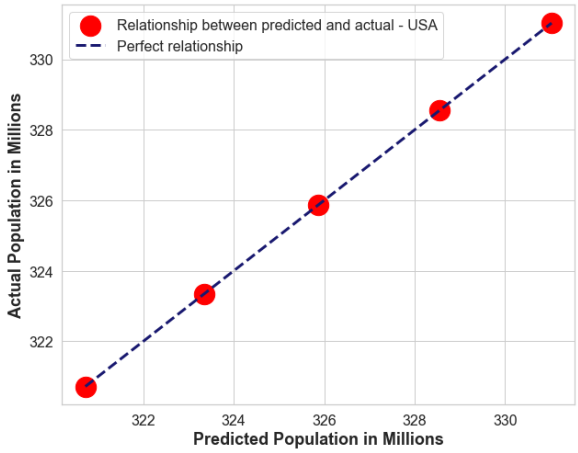
(a) Total population of India



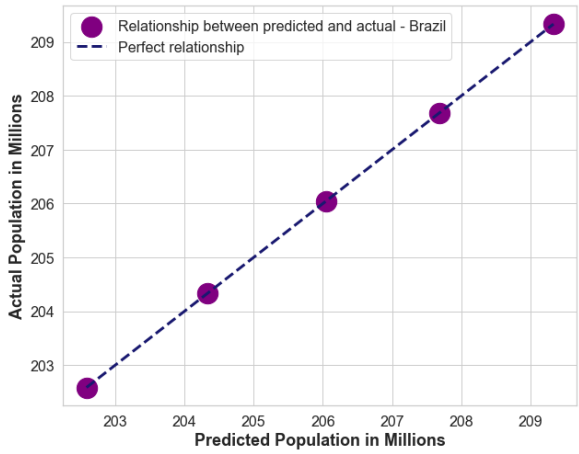
(b) Total population of China



(c) Total population of the UK



(d) Total population of the USA



(e) Total population of Brazil

Figure 16: A plot of the actual total population (training data from 2003-2013, testing data from 2014-2018) against the predicted total population using the linear regression model for all five countries from 2013-2018.

Table 3 shows the values of the performance metrics in predicting the total population using the linear regression model. The resilience of the model is proven by the R^2 value which approximated unity across all countries, implying a nearly perfect fit. Concurrently, supplementary metrics such as MAE and MSE as shown in table 3 were very small, confirming the accuracy

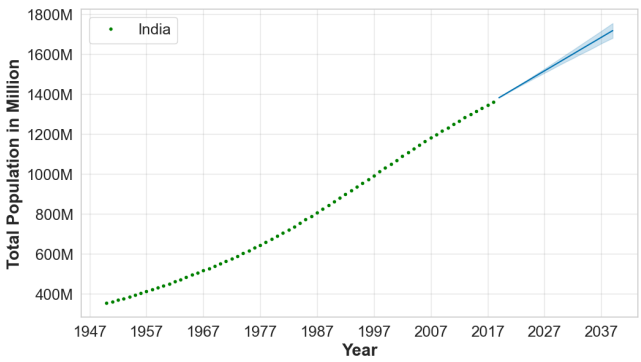
of the model.

Table 3: Performance metrics implemented for predicting the total population for the countries under consideration from 2014-2018 using the linear regression model.

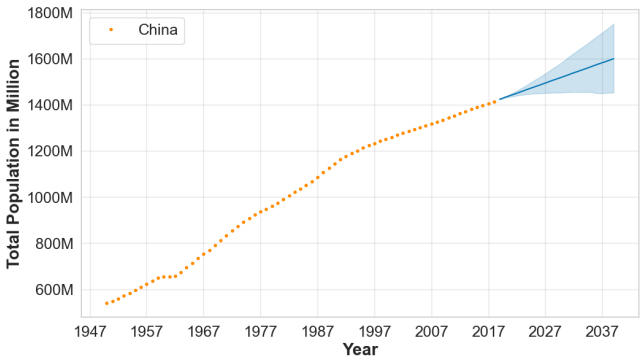
Errors in total population			
Countries	MSE	MAE	R^2
Brazil	0.001	0.023	0.9999
China	154.593	10.336	0.9985
India	0.034	0.144	0.9996
United Kingdom	0.001	0.019	0.9998
United States of America	0.000	0.002	0.9999

In addition to this, the average error calculated by averaging the errors of the five countries was close to zero. The linear regression model is precise and can be used for forecasting future populations and analyzing historical populations. This technique primarily generates linear estimates, which may not adequately capture fluctuations or declines in population growth rates, resulting from policy implementations or population control measures in future. In order to accommodate non-linear patterns in population data, an advanced time series model called the prophet model is used for forecasting the total population of our five countries for a period of 20 years from 2019-2039.

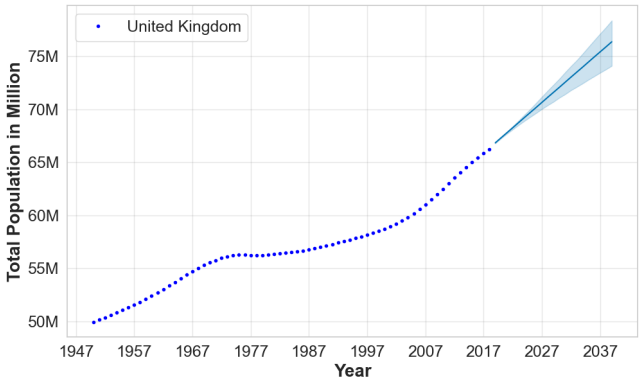
Figure 17 depicts the forecasted total population of all five countries for the years 2019-2039 using the prophet model. In figure 17, the dots represents the actual values from 1950 to 2018 while the solid line shows the forecasted values from 2019 to 2039. The shaded region surrounding the forecasting line is an indicator of the uncertainty ranges relevant to the forecasts. This defined area provides an estimate of the range within which future values are expected to evolve, based on the underlying assumptions of the specified statistical models and the inherent variability present in the data set. On analyzing this uncertainty of the forecasted population trajectories, it is observed that India and Brazil exhibit the narrowest range of certainty intervals among the five countries under study, denoting minimal variations in their forecasted populations.



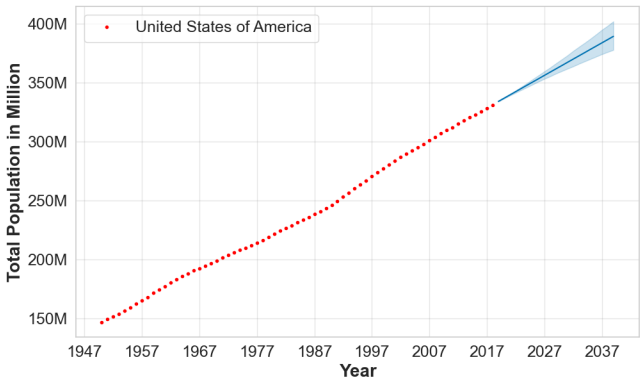
(a) Forecasted Population of India (2019-2039)



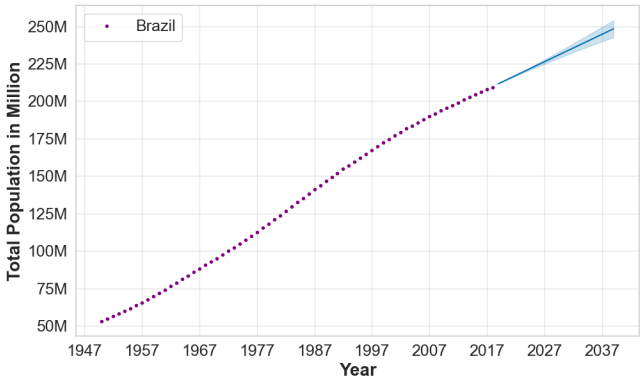
(b) Forecasted Population of China (2019-2039)



(c) Forecasted Population of UK (2019-2039)



(d) Forecasted Population of USA (2019-2039)



(e) Forecasted Population of Brazil(2019-2039)

Figure 17: A plot of the forecasted total population (training data from 1950-2018) against the corresponding years using the prophet model for all five countries from 2019-2039.

On the other hand, the UK and the USA manifest moderate trusted intervals, while China displays the widest, indicating greater fluctuations in its future population estimates. In addition to utilizing the Prophet model, Exponential Smoothing is also executed for the projected total population for the five countries. However, it is imperative to note that the average R^2 value yielded by this method is approximately 95 percent, which is less than that of the Prophet model and Linear Regression, both of which exhibited an average R^2 value of 99 percent.

Chapter 5

5.1 Conclusion

Addressing the significance of understanding population growth, this dissertation meticulously examined the demographic trajectories of five pivotal nations: India, China, the UK, the USA, and Brazil. The primary aim is to identify the variables influencing their population growth and to understand the broader implications of these individual trends on global demographics. Central to this demographic examination is identifying the determinants of population growth within these nations which we have initiated in the literature review. Birth rates emerge as a significant factor followed by death rates and migratory patterns. Distinct patterns emerge across the five countries. India and Brazil display pronounced youthful demographics, while the older population of China might be due to its former stringent family planning policy of one child. The UK indicates an aging population, raising concerns about possible dependency challenges. The USA, with its broad age spectrum, seems to benefit from diverse age dynamics. Gender imbalances are particularly marked in China and the UK, with a historical decline in the male populations of the UK and the USA around 1922, likely due to the consequence of the World War I. Birth rates further explicate these demographic trends. The one-child policy of China, enforced between 1980 and 2016, significantly impacted its demographics and indirectly propelled India to its position as the most populated nation by 2023. Meanwhile, the stable birth rates of the UK align with its older population trend. Migration patterns also offer valuable insights. The economic and cultural appeal of the USA and the UK attract diverse global migrants, while Brazil shows a fluctuating pattern since 1950. China and India are observed as net human capital exports. The CNMR values for the USA represents the vital role of migration in its demographic configuration. The demographic shifts of China, denoted by the policy-induced variations, highlight the influence of governmental decisions. The population growth transition of Brazil from 3 percentage to 0.5 percentage growth highlights the effect of the interference of socio-economic reforms. As a result, it is apparent that policy implications can balance population growth by implementing population control measures and migrant restrictions. These approaches can help safeguard the current population by assuring sustainability. The data set analysis in chapter 2 can examined using the ML techniques in chapter 3, namely linear regression,ARIMA,GBR and exponential smoothing, to predict CBR. Additionally, methodologies such as linear regression and Prophet, are used to anticipate future demographic patterns. The results indicates that linear regression, a fundamental machine learning algorithm, is optimally suited for the predictions of CDR,CBR and total population. For data sets exhibiting non-linearity, the GBR is found effective for

finding the dependency of the other features of the data set in predicting both CBR and CDR. Conversely, the Prophet method emerged as the superior choice for forecasting the total population, ensuring high precision. A fine correlation was observed among all features in the case of CBR. It is also noteworthy that a decrease in CBR was observed across all five countries. The accessibility of healthcare, shifts in lifestyle, and governmental policies, like the one-child policy of China or the post-Brexit migration policies of the UK, desperately influence these rates and, by extension, demographic patterns. It is critical to understand that variations in these rates, whether increasing or decreasing, have a substantial impact on population structures. A surge in birth rates without an analogous rise in death rates may yield a predominantly younger population. On the other hand, a nation witnessing declining births but consistent death rates may lead to an aging demographic. Such shifts bear further consequences for socio-economic infrastructures, from healthcare to employment. Undoubtedly, the aggregate impact of these individual country trajectories on global demographics is significant. The demographic dynamics of India and China, owing to their large size, hold the potential to significantly alter global metrics. Alterations in their demographic patterns have consequences beyond their borders, influencing global trade, economics, and politics. The expected exponential rise of the world population in the coming decades demonstrates the need to understand these processes, particularly when revealing the responsibilities of geo-politically and economically powerful nations such as the USA, the UK, and Brazil. On analysing the contribution of each countries under observation to the world population from 1950 to 2018. India's rapid population increase, along with its distinct socio-cultural pattern, has far-reaching global implications, ranging from migratory patterns to commercial exchanges. In contrast, China's demographic trajectory, affected by its one-child policy, has resulted in significant population crises. Despite differences in growth rates, the USA, Brazil, and the UK play critical roles in defining the global demographic paradigm due to their geopolitical relevance, cultural inheritance, and economic contributions. In summary, the combined demographic contributions of these nations provide an overview of global demographic evolution, influencing economic trajectories, global policies, migratory preferences, and concerns such as sustainability. These nations, with different histories and cultures, drive the global demographic change, illustrating that it is a confluence of numerous variables, ranging from historical footprints to policy decisions. By exploring thoroughly into these complexities, our research aims to provide an extensive awareness of global population dynamics, providing as a guide for politicians, academics, and industry as they navigate the complexity of a dynamic global demographic context.

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Appendix

	(a) Crude Birth Rate (CBR)					(b) Crude Death Rate (CDR)				
Country	Min	Q1	Median	Q3	Max	Min	Q1	Median	Q3	Max
Brazil	12.48	17.34	27.33	36.14	46.43	6.11	6.49	7.80	11.06	17.32
China	7.48	13.29	21.88	35.29	48.99	6.04	6.53	7.19	12.65	24.83
India	16.14	23.94	33.68	39.79	44.24	6.56	8.19	11.52	17.51	22.39
UK	10.02	12.03	12.95	15.74	18.85	8.72	9.72	11.36	11.75	12.57
USA	10.94	14.07	15.46	17.44	23.97	7.91	8.45	8.69	9.36	9.74

(c) Population Growth Rate (%)

Country	Min	Q1	Median	Q3	Max
Brazil	0.45	1.10	1.97	2.50	3.09
China	-0.03	0.65	1.33	1.95	3.26
India	0.68	1.56	2.16	2.24	2.31
UK	-0.04	0.23	0.41	0.59	0.84
USA	0.30	0.95	1.04	1.34	2.08

	(d) Life Expectancy (Years)					(e) Median Age (Years)				
Country	Min	Q1	Median	Q3	Max	Min	Q1	Median	Q3	Max
Brazil	48.12	56.75	65.08	71.75	76.18	16.75	17.45	20.76	26.11	33.62
China	39.70	55.74	67.29	74.11	78.79	17.96	20.00	22.52	31.73	38.98
India	41.72	47.86	57.31	65.00	72.03	18.23	19.04	19.94	22.60	28.25
UK	68.24	71.88	75.15	79.11	82.31	32.81	33.94	34.52	37.68	40.06
USA	68.06	70.52	74.83	77.41	79.74	27.24	28.78	30.97	35.08	38.10

Table 4: Descriptive Statistics of Demographic Metrics for Selected Countries (1950-2023)